M2 Quantitative Finance · Machine Learning in Finance

Deep Hedging

Replication and Analysis of Neural Network-Based
Derivatives Hedging Under Market Frictions

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Abstract

This report presents a full replication and extended analysis of the Deep Hedging framework introduced by Buehler et al. (2019), in which neural networks are trained to hedge derivatives portfolios in the presence of market frictions. We implement the complete pipeline from scratch in PyTorch, covering market simulation, option pricing, network architecture, and training under two risk measures: Conditional Value-at-Risk (CVaR) and the entropic risk measure. Experiments are conducted in two market environments, Black & Scholes and Heston stochastic volatility, allowing a systematic comparison against classical delta hedging baselines.

Our results confirm the central claim of the paper: once proportional transaction costs are introduced, tail-focused deep hedging strategies ($\alpha = 0.95, 0.99$) substantially outperform the classical model hedge, reducing CVaR(95%) by up to 24% in Black & Scholes and 20% in Heston, while reducing average transaction costs by 42% and 62% respectively. We further show that the CVaR objective consistently outperforms the entropic objective for tail risk management: in the Heston setting, the entropic model with $\lambda = 1$ proves insufficiently risk-averse and actually produces worse tail metrics than the classical baseline, highlighting the importance of objective choice in stochastic volatility environments. A hyperparameter sensitivity analysis reveals that the Heston environment benefits from significantly larger network capacity than Black & Scholes, reflecting the increased complexity of two-asset stochastic volatility hedging.

Finally, we numerically verify the $O(\epsilon^{2/3})$ price asymptotics for the exponential utility indifference price under proportional transaction costs. In Black & Scholes we recover an empirical slope of 0.69, in close agreement with the theoretical value of 2/3. In the Heston two-asset setting we recover an empirical slope of 0.63, confirming the qualitative $O(\epsilon^{2/3})$ behaviour in a multi-asset stochastic volatility model.

Keywords: Deep Hedging, Reinforcement Learning, Derivatives Hedging, Transaction Costs, CVaR, Entropic Risk Measure, Heston Model, Neural Networks, Market Frictions

Table of Contents

1 Introduction **1**

2 Theoretical Framework **2**

2.1 Discrete-Time Market with Frictions . . . 2

2.1.1 Market Setup and Trading Strategies . . . 2

2.1.2 Transaction Costs and Terminal P&L . . . 2

2.2 Pricing and Hedging under Convex Risk Measures . . . 3

2.2.1 Convex Risk Measures . . . 3

2.2.2 The Hedging Optimisation Problem . . . 3

2.2.3 Indifference Pricing . . . 3

2.2.4 Conditional Value-at-Risk and Expected Shortfall . . . 4

2.2.5 Entropic Risk Measure and Exponential Utility . . . 4

2.3 Deep Hedging: Neural Network Parametrisation . . . 5

2.3.1 Universal Approximation . . . 5

2.3.2 Network Architecture . . . 5

2.3.3 Semi-Recurrent Parametrisation and Input Features . . . 6

2.3.4 Convergence Guarantee . . . 7

2.4 Training Procedure . . . 7

2.4.1 CVaR Optimisation via the Rockafellar-Uryasev Formula . . . 7

2.4.2 Entropic Optimisation . . . 7

2.4.3 Optimiser, Scheduler and Gradient Clipping . . . 8

2.4.4 On-Policy Path Generation . . . 8

2.5 Evaluation Protocol and Metrics . . . 8

3 Market Simulation and Classical Baselines **9**

3.1 Black & Scholes . . . 10

3.1.1 Path Generation . . . 10

3.1.2 Analytical Pricer . . . 10

3.1.3 Classical Baseline . . . 11

3.2 Heston Stochastic Volatility . . . 11

3.2.1 Model and Parameters . . . 11

3.2.2 Path Generation: Exact CIR Sampling . . . 12

3.2.3 Path Generation: Simplified Broadie-Kaya Scheme . . . 12

3.2.4 Variance Swap as Second Hedging Instrument . . . 13

3.2.5 Heston Option Pricer via Fourier Inversion . . . 14

3.2.6 Numerical Greeks and Classical Heston Baseline . . . 14

4 Numerical Experiments **15**

4.1 Black & Scholes 15
4.1.1 Black & Scholes Without Transaction Costs 16
4.1.2 Black & Scholes With Transaction Costs 19
4.1.3 Hyperparameter Sensitivity: Black & Scholes 22
4.1.4 Black & Scholes: Entropic Risk Measure 24
4.2 Heston 26
4.2.1 Heston Without Transaction Costs 27
4.2.2 Heston With Transaction Costs 29
4.2.3 Heston: Entropic Risk Measure 33
4.2.4 Heston CVaR vs Transaction Cost Level 35
4.2.5 Hyperparameter Sensitivity: Heston 36
4.3 Price Asymptotics Under Proportional Transaction Costs 39
4.3.1 Black & Scholes 39
4.3.2 Heston 40

5 Conclusion **42**

References **45**

1 Introduction

The problem of hedging a portfolio of derivatives is central to risk management in the financial securities industry. In idealised frictionless markets, mathematical finance provides a tractable solution: under the Black & Scholes model, a derivative can be replicated perfectly by continuously trading the underlying asset according to the analytical delta. Real markets, however, are subject to frictions. Every transaction incurs a cost, liquidity is finite, and trading desks operate under risk and capital constraints. These frictions fundamentally alter the hedging problem: the optimal strategy is no longer the frictionless delta, and pricing and risk become non-linear quantities depending on the overall book. Despite this, the industry has long relied on classical model-based greeks as a practical approximation, largely due to the absence of scalable model-free alternatives. Buehler et al. (2019) propose a fundamentally different approach: rather than computing greeks from a pricing model and adjusting for frictions after the fact, they parametrise the hedging strategy directly as a neural network and train it end-to-end to minimise a convex risk measure of the terminal profit and loss, with transaction costs explicitly included in the objective. This Deep Hedging framework requires no equivalent martingale measure and no ability to compute greeks. Its computational cost depends primarily on the number of hedging instruments rather than the size of the portfolio, making it scalable to realistic trading books.

This report presents a full replication and extended analysis of the Deep Hedging framework, implemented from scratch in PyTorch. Section 2 presents the theoretical framework, covering the discrete-time market with frictions, convex risk measures, the neural network parametrisation, and the training procedure. Section 3 describes the market simulators and classical baselines used throughout. Section 4 presents the numerical experiments in two environments: Black & Scholes, which serves as a validation setting where the analytical delta provides a strong benchmark, and Heston stochastic volatility, where a variance swap is used as a second hedging instrument. Within each environment we study the effect of transaction costs, compare the CVaR and entropic objectives, analyse hyperparameter sensitivity, and verify the theoretical $O(\epsilon^{2/3})$ price asymptotics for the exponential utility indifference price under proportional transaction costs. Section 5 summarises the findings and discusses practical implications.

Our results confirm the central claim of Buehler et al. (2019): once proportional transaction costs are introduced, tail-focused deep hedging strategies substantially outperform classical model-based hedging, reducing CVaR(95%) by up to 24% and average transaction costs by 42% in Black & Scholes, and by 20% and 62% respectively in Heston. We further find that the choice of risk objective matters significantly: the entropic model underperforms the CVaR objective on tail metrics in both environments, and fails to improve upon the classical baseline in the Heston setting.

2 Theoretical Framework

2.1 Discrete-Time Market with Frictions

2.1.1 Market Setup and Trading Strategies

We consider a discrete-time financial market with finite time horizon T and n equally spaced trading dates $0 = t_0 < t_1 < \cdots < t_n = T$, with time step $\Delta t = T/n$. The market contains d hedging instruments whose mid-prices are described by an $\mathbb{R}^d$-valued adapted stochastic process $S = (S_k)_{k=0,\ldots,n}$. Our liability is an $\mathcal{F}_T$-measurable random variable Z, representing the terminal payoff of the derivatives portfolio to be hedged.

At each rebalancing date t_k, the agent observes a market information vector $I_k \in \mathbb{R}^r$ comprising mid-prices, implied volatilities, and any other relevant features. The filtration $\mathbb{F} = (\mathcal{F}_k)_{k=0,\ldots,n}$ is generated by $I = (I_k)_{k=0,\ldots,n}$.

A trading strategy is an $\mathbb{R}^d$-valued $\mathbb{F}$-adapted process $\delta = (\delta_k)_{k=0,\ldots,n-1}$, where δ_k^i denotes the holdings in the i-th instrument at time t_k. We denote by $\mathcal{H}$ the set of all admissible strategies, which may be subject to liquidity or position constraints.

In our implementation, we use $T = 30/365$, $n = 30$ daily rebalancing steps, $d = 1$ in the Black & Scholes environment, and $d = 2$ in the Heston environment, matching the parameters of Buehler et al. (2019) exactly.

2.1.2 Transaction Costs and Terminal P&L

All trading is self-financed. If the agent receives a premium p_0 at inception and follows strategy δ, the terminal profit and loss is defined as (Buehler et al., 2019):

$$\mathrm{PnL}_T(Z, p_0, \delta) = p_0 + \sum_{k=0}^{n-1} \delta_k \cdot (S_{k+1} - S_k) - C_T(\delta) - (S_T^1 - K)^+ \tag{1}$$

where the total proportional transaction cost is:

$$C_T(\delta) = \epsilon \cdot S_0 |\delta_0| + \sum_{k=1}^{n-1} \epsilon \cdot S_k |\delta_k - \delta_{k-1}| \tag{2}$$

Here $\epsilon > 0$ is the proportional cost rate, set to $\epsilon = 0.005$ in all experiments with transaction costs. This structure penalises both the initial entry into the hedge and all subsequent rebalancing trades, making it costly to track the frictionless delta closely. At maturity, liquidation is assumed to be cost-free, consistent with the assumption of Buehler et al. (2019). In the two-asset Heston setting, the same cost structure applies independently to

both the stock leg and the variance swap leg, with each instrument's mid-price used as the notional for cost computation.

2.2 Pricing and Hedging under Convex Risk Measures

2.2.1 Convex Risk Measures

Definition 2.1 (Convex Risk Measure)**.** A functional $\rho : \mathcal{X} \to \mathbb{R}$ is called a convex risk measure if it satisfies:

(i) **Monotonicity:** if $X_1 \geq X_2$ then $\rho(X_1) \leq \rho(X_2)$. A more favourable position requires less risk capital.

(ii) **Convexity:** $\rho(\lambda X_1 + (1-\lambda)X_2) \leq \lambda\rho(X_1) + (1-\lambda)\rho(X_2)$ for all $\lambda \in [0,1]$. Diversification between two positions never increases risk.

(iii) **Cash-invariance:** $\rho(X + c) = \rho(X) - c$ for all $c \in \mathbb{R}$. Adding cash c to a position reduces the required capital injection by exactly c.

The cash-invariance property implies that $\rho(X)$ represents the minimum cash amount that must be added to position X to make it acceptable, in the sense that $\rho(X + \rho(X)) = 0$.

2.2.2 The Hedging Optimisation Problem

Given a convex risk measure ρ, the optimal hedging problem is defined as:

$$\pi(X) := \inf_{\delta \in \mathcal{H}} \rho\big(X + (\delta \cdot S)_T - C_T(\delta)\big) \tag{3}$$

This represents the minimum residual risk achievable by optimally hedging position X over all admissible strategies. Under convexity of $\mathcal{H}$ and C_T, the functional π is itself a convex risk measure (Buehler et al., 2019).

2.2.3 Indifference Pricing

The indifference price $p(Z)$ of a contingent claim Z is the cash amount that makes the agent indifferent between writing Z and not doing so:

$$p(Z) := \pi(-Z) - \pi(0) \tag{4}$$

By cash-invariance, this is equivalent to solving $\pi(-Z + p_0) = \pi(0)$ for p_0. In a complete frictionless market, $p(Z)$ reduces to the unique replication cost. The decomposition in equa-

tion (4) is central to the price asymptotics experiment in Section 4: for each transaction cost level ϵ, we train two separate models, one with the option liability included to approximate $\pi(-Z;\epsilon)$ and one without to approximate $\pi(0;\epsilon)$, and compute $p_\epsilon = \pi(-Z;\epsilon) - \pi(0;\epsilon)$.

2.2.4 Conditional Value-at-Risk and Expected Shortfall

The Conditional Value-at-Risk at level $\alpha \in (0,1)$, also called Expected Shortfall, is defined as:

$$\mathrm{CVaR}_\alpha(X) := \frac{1}{1-\alpha}\int_0^{1-\alpha} \mathrm{VaR}_\gamma(X)\, d\gamma \tag{5}$$

where $\mathrm{VaR}_\gamma(X) = \inf\{m \in \mathbb{R} : \mathbb{P}(X < -m) \leq \gamma\}$. It represents the expected loss in the worst $(1-\alpha)$ fraction of scenarios and is a coherent risk measure. Higher values of α correspond to greater risk aversion, focusing optimisation on increasingly extreme tail losses.

For gradient-based optimisation, we use the Rockafellar-Uryasev representation (Rockafellar and Uryasev, 2000):

$$\mathrm{CVaR}_\alpha(L) = \min_{v\in\mathbb{R}} \left\{ v + \frac{1}{1-\alpha}\,\mathbb{E}[\max(L-v,0)] \right\} \tag{6}$$

where $L = -\mathrm{PnL}$ denotes the loss. This representation is differentiable with respect to both the network parameters θ and the scalar threshold v, which converges to VaR_α at the optimum. Both θ and v are jointly optimised during training. We study three risk levels: $\alpha \in \{0.50, 0.95, 0.99\}$, where $\alpha = 0.50$ corresponds to a median-focused objective and $\alpha = 0.99$ concentrates optimisation on the extreme tail.

2.2.5 Entropic Risk Measure and Exponential Utility

For risk-aversion parameter $\lambda > 0$, the entropic risk measure is:

$$\rho_\lambda(X) = \frac{1}{\lambda}\log \mathbb{E}\big[e^{-\lambda X}\big] \tag{7}$$

This is the convex risk measure associated with the exponential utility function $U(x) = -e^{-\lambda x}$, and corresponds to the exponential utility indifference pricing framework studied in Buehler et al. (2019). As $\lambda \to 0$, the measure converges to $-\mathbb{E}[X]$, recovering the risk-neutral objective. Higher values of λ penalise large losses more heavily. We use $\lambda = 1$ throughout, matching the paper's choice.

In our implementation, the entropic objective is computed using the numerically stable

log-sum-exp identity:

$$\frac{1}{\lambda}\log\mathbb{E}\left[e^{-\lambda X}\right] = \frac{1}{\lambda}\left(\log\sum_{m=1}^{N} e^{-\lambda X_m} - \log N\right) \tag{8}$$

which avoids floating-point overflow that would arise from naively computing the expectation of the exponential directly. Unlike the CVaR objective, the entropic optimisation requires no auxiliary parameter: only the network weights θ are optimised.

2.3 Deep Hedging: Neural Network Parametrisation

2.3.1 Universal Approximation

The theoretical foundation for using neural networks to approximate optimal hedging strategies rests on the universal approximation theorem (Hornik, 1991): for any bounded non-constant activation function σ, the class of feedforward neural networks is dense in $L^p(\mathbb{R}^{d_0}, \mu)$ for any finite measure μ and $1 \leq p < \infty$. This guarantees that any measurable hedging strategy can be approximated arbitrarily well by a sufficiently large neural network, providing the theoretical justification for the parametrisation described below.

2.3.2 Network Architecture

Each individual network in our implementation is a fully connected feedforward neural network with the following structure. For each hidden layer of width h, the computation proceeds as:

1. a linear transformation $W \in \mathbb{R}^{h \times h_{\text{prev}}}$,
2. a ReLU activation $\sigma(x) = \max(x, 0)$ applied elementwise.

The final layer is a linear transformation to the output dimension with no activation. All weight matrices are initialised with Xavier uniform initialisation and all bias vectors are initialised to zero. Batch normalisation is disabled throughout: although the original paper applies it, we found empirically that it destabilised training under on-policy simulation, leading to higher out-of-sample CVaR values, and therefore disabled it.

The hidden layer widths follow the prescription of Buehler et al. (2019): $N_1 = N_2 = d + 15$, giving $[16, 16]$ for Black & Scholes ($d = 1$) and $[17, 17]$ for Heston ($d = 2$). The output dimension is d, matching the number of hedging instruments.

2.3.3 Semi-Recurrent Parametrisation and Input Features

The optimal hedging strategy at time t_k depends on the current market information I_k and on the previous position δ_{k-1}, since transaction costs make the cost of rebalancing depend on the current holdings. Following Buehler et al. (2019), we parametrise the strategy as:

$$\delta_k^\theta = F_k^\theta(I_k, \delta_{k-1}^\theta) \tag{9}$$

where F_k^θ is a feedforward neural network. We implement two policy variants:

Per-step policy. A separate network F_k^θ is maintained for each rebalancing date t_k, stored in a ModuleList of $n = 30$ networks. Parameters are not shared across time steps, allowing each network to specialise to its own position in the hedging horizon. This is the primary architecture of Buehler et al. (2019) and the one used in all main experiments. Since each network is dedicated to a fixed time step, the time-to-maturity $\tau_k = T - t_k$ is not included as an input feature.

Shared policy. A single network is reused at every rebalancing date, with the time-to-maturity τ_k added as an explicit input feature to allow the network to distinguish time steps. This reduces the parameter count by a factor of n at the cost of less expressive time-dependent hedging policies.[1]

The input feature vector at time t_k is constructed as follows. For the per-step policy:

- **Black & Scholes:** $x_k = (\log(S_k/S_0),\ \delta_{k-1}) \in \mathbb{R}^2$.
- **Heston:** $x_k = (\log(S_k^1/S_0),\ V_k/V_0,\ \delta_{k-1}^{(1)},\ \tilde{\delta}_{k-1}^{(2)}) \in \mathbb{R}^4$.

For the shared policy, τ_k is appended to each feature vector, increasing the input dimension by one. The log-spot $\log(S_k/S_0)$ is used instead of the raw price to produce a stationary, approximately centred input. The variance V_k is normalised by $V_0 = 0.04$ to bring it to the same order of magnitude as the log-spot.

Variance-swap normalisation. The raw variance swap price satisfies $S_0^2 = L(0, V_0) \approx 0.003$, which is approximately 30,000 times smaller than the stock price $S_0^1 = 100$. Without correction, the two output heads of the Heston network would operate on completely different scales, making joint optimisation unstable. We therefore apply the normalisation

[1] The shared policy architecture was implemented and remains available as a configurable option in the codebase, but all experiments reported in this paper use the per-step policy exclusively, as running and reporting results for both architectures was beyond the scope of this assignment. The shared policy nonetheless produces interesting results and can be easily activated in the codebase.

$\tilde{S}^2 = S^2 \cdot S_0/L_0$, where $L_0 = L(0, V_0)$ is the initial variance swap price computed from equation (5.4) of Buehler et al. (2019). This rescaling is applied consistently during training, evaluation, and transaction cost computation, so that the network's δ_2 output and the associated costs remain on a comparable scale to the stock leg throughout.

2.3.4 Convergence Guarantee

Proposition 2.1 (Buehler et al. (2019), Proposition 4.9)**.** *Let $\mathcal{H}_M$ denote the set of trading strategies parametrised by neural networks with at most M non-zero weights, and let π^M be the corresponding approximate risk. Then:*

$$\lim_{M\to\infty} \pi^M(X) = \pi(X) \quad \textit{for all } X \in \mathcal{X} \tag{10}$$

This result guarantees that the neural network parametrisation introduces no approximation bias in the limit, and that the deep hedging price converges to the true indifference price as network capacity grows. It also justifies the hyperparameter sensitivity study in Section 4, where we verify empirically that increasing network width yields measurable improvements in out-of-sample CVaR, with the rate of improvement differing substantially between the Black & Scholes and Heston environments.

2.4 Training Procedure

2.4.1 CVaR Optimisation via the Rockafellar-Uryasev Formula

For the CVaR objective, the optimisation problem over the network parameters θ and the auxiliary VaR threshold v is:

$$\min_{\theta,\, v} \left\{ v + \frac{1}{1-\alpha}\, \mathbb{E}\big[\max(-\mathrm{PnL}_T^\theta - v,\, 0)\big] \right\} \tag{11}$$

The loss decomposes as a sum over paths, enabling efficient minibatch stochastic gradient descent. Both θ and v are updated simultaneously by the Adam optimiser at each training step.

2.4.2 Entropic Optimisation

For the entropic objective, the optimisation problem is:

$$\min_{\theta}\ \frac{1}{\lambda} \log \mathbb{E}\big[e^{-\lambda\, \mathrm{PnL}_T^\theta}\big] \tag{12}$$

implemented via the log-sum-exp identity described in Section 2. Only the network weights θ are optimised; no auxiliary parameter is required.

2.4.3 Optimiser, Scheduler and Gradient Clipping

Both objectives are minimised using the Adam optimiser, with learning rate $\eta = 0.005$. A cosine annealing learning rate scheduler decays the learning rate smoothly from 0.005 to 10^{-5} over the full training horizon, allowing coarse parameter updates in early epochs and fine-grained adjustments as training progresses.

To prevent gradient instability, which is particularly pronounced in the Heston setting due to the high variance of the two-asset on-policy training distribution, we apply gradient clipping with a maximum ℓ_2 norm of 1.0 for Black & Scholes and 5.0 for Heston. The stricter clipping in the Black & Scholes setting reflects the simpler and more stable gradient landscape under constant volatility.

Training uses 500 epochs for Black & Scholes and 1500 epochs for Heston, with a batch size of 4096 throughout. The larger epoch budget for Heston reflects the increased complexity of two-asset stochastic volatility hedging: as discussed in Section 4, models stopped early exhibit meaningfully worse tail metrics on the test set despite the training objective appearing flat, suggesting that the network continues to refine its hedging policy in later epochs.

2.4.4 On-Policy Path Generation

At each training epoch, a fresh batch of market paths is simulated from scratch rather than drawing from a fixed pre-generated dataset. This on-policy approach ensures that the network is always trained on paths consistent with its current induced distribution of hedge positions, and prevents overfitting to a fixed training set. It also means that the effective training sample size is epochs $\times$ batch size: $500 \times 4096 = 2{,}048{,}000$ paths for Black & Scholes and $1500 \times 4096 = 6{,}144{,}000$ paths for Heston.

2.5 Evaluation Protocol and Metrics

After training, all policies are evaluated on a fixed out-of-sample test set of 50,000 paths generated with a fixed random seed, ensuring that the analytical baseline, the CVaR model, and the entropic model are all compared on identical market scenarios. The following metrics are computed from the resulting terminal P&L distribution:

All metrics are computed using the $\lceil \alpha N \rceil$ quantile cutoff to ensure consistency between the CVaR computed during training via the Rockafellar-Uryasev formula and the CVaR

Metric	Definition
Mean P&L	$\mathbb{E}[\mathrm{PnL}_T]$
Std P&L	$\mathrm{std}(\mathrm{PnL}_T)$
VaR(α)	q_α of the loss distribution $-\mathrm{PnL}_T$
CVaR(α)	$\mathbb{E}[-\mathrm{PnL}_T \mid -\mathrm{PnL}_T > \mathrm{VaR}_\alpha]$
Turnover	$\mathbb{E}\left[\lvert\delta_0\rvert + \sum_{k=1}^{n-1} \lvert\delta_k - \delta_{k-1}\rvert\right]$
Avg Transaction Cost	$\mathbb{E}[C_T(\delta)]$

Table 1: Risk metrics computed on the out-of-sample test set. VaR and CVaR are reported at both $\alpha = 95\%$ and $\alpha = 99\%$.

reported in the evaluation tables.

3 Market Simulation and Classical Baselines

This section describes the two market environments used throughout the experiments, the option pricing models used to compute premiums and classical greeks, and the construction of the classical hedging baselines against which the deep hedge is compared. Black & Scholes serves as a validation setting where the analytical delta provides a strong and theoretically optimal benchmark, while Heston stochastic volatility is the primary setting of interest, where the variance swap is used as a second hedging instrument to hedge volatility risk. Table 2 summarises the common parameters used across both environments.

Parameter	Description	BS	Heston
S_0, K	Initial price, strike	100	100
T, n	Maturity, rebalancing steps	30/365, 30	30/365, 30
r	Risk-free rate	0	0
σ	Constant volatility	0.20	—
κ	Mean-reversion speed	—	1
θ	Long-run variance	—	0.04
σ_v	Volatility of variance	—	2
ρ	Spot-variance correlation	—	-0.70
V_0	Initial variance	—	0.04
p_0	Call premium	2.2871	1.6444
ϵ	Proportional transaction cost	0.005	

Table 2: Market and simulation parameters used in all experiments, matching the calibration of Buehler et al. (2019) exactly.

Both test sets consist of $N = 50{,}000$ paths generated with fixed random seeds (NumPy seed 42, PyTorch seed 42). All compared strategies, the analytical or numerical classical baseline and all deep hedge variants, are evaluated on identical market scenarios, ensuring that performance differences are not attributable to sampling variability.

3.1 Black & Scholes

3.1.1 Path Generation

Under the Black & Scholes model, the asset price S_t follows geometric Brownian motion with constant drift r and constant volatility σ. The SDE admits an exact closed-form solution, so the transition over one time step $\Delta t = T/n$ is given exactly by:

$$S_{t+\Delta t} = S_t \exp\Big(\big(r - \tfrac{1}{2}\sigma^2\big)\,\Delta t + \sigma\sqrt{\Delta t}\,Z\Big), \quad Z \sim \mathcal{N}(0,1) \tag{13}$$

This is not an Euler-Maruyama approximation but the exact transition of geometric Brownian motion, obtained by applying Itô's formula to $\log S_t$. No discretisation error is introduced by the path generator itself; any residual hedging error in the frictionless experiment therefore arises exclusively from the discrete rebalancing schedule.

In our implementation, all $n \times N$ standard normal shocks are generated simultaneously in a single vectorised GPU operation, and the asset price path for each of the N scenarios is obtained by exponentiating the cumulative sum of log-returns. The complete path tensor of shape $(n+1) \times N$ is constructed without any Python-level loop, making the simulator extremely efficient on GPU.

3.1.2 Analytical Pricer

The Black & Scholes European call price and its analytical delta are:

$$C(S,K,\tau) = S\,\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad \Delta(S,K,\tau) = \Phi(d_1) \tag{14}$$

$$d_1 = \frac{\log(S/K) + \big(r + \frac{1}{2}\sigma^2\big)\,\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau} \tag{15}$$

where Φ denotes the standard normal cumulative distribution function and $\tau = T - t_k$ is the time to maturity at rebalancing date t_k. In our implementation, Φ is evaluated via the identity $\Phi(x) = \frac{1}{2}(1 + \mathrm{erf}(x/\sqrt{2}))$, which is numerically stable and available as a native PyTorch operation on GPU, allowing the delta to be computed for all N paths simultaneously without leaving the GPU. The time-to-maturity τ is clamped to a minimum of 10^{-6} to prevent division by zero at expiry. At $\tau = 0$, the delta degenerates to the indicator $\mathbf{1}_{S>K}$, which is handled explicitly. The at-the-money call premium is computed analytically as $p_0 = 2.2871$.

3.1.3 Classical Baseline

The analytical delta hedge computes $\Delta(S_{t_k}, K, T - t_k)$ at each of the $n = 30$ rebalancing dates and applies the terminal P&L formula of equation (1) with proportional transaction costs. By construction, this strategy is cost-oblivious: it rebalances to the frictionless optimal delta at every step without any regard for the transaction costs incurred. In the frictionless setting it is the variance-minimising strategy under geometric Brownian motion and therefore represents the strongest achievable benchmark. Once transaction costs are introduced, however, its inability to account for rebalancing costs becomes a structural limitation, which is precisely the regime where deep hedging is expected to provide measurable improvements.

3.2 Heston Stochastic Volatility

3.2.1 Model and Parameters

The Heston model (Heston, 1993) describes the joint evolution of the asset price S_t^1 and its instantaneous variance V_t through the coupled system of stochastic differential equations:

$$dS_t^1 = \sqrt{V_t}\, S_t^1\, dB_t \tag{16}$$

$$dV_t = \kappa(\theta - V_t)\, dt + \sigma_v \sqrt{V_t}\, dW_t \tag{17}$$

where $\langle dB_t, dW_t \rangle = \rho\, dt$ and we set $r = 0$ for simplicity. The variance process is a Cox-Ingersoll-Ross (CIR) process with mean-reversion speed κ, long-run mean θ, and volatility-of-volatility σ_v. The Feller condition $2\kappa\theta > \sigma_v^2$ is a sufficient condition for the variance process to remain strictly positive almost surely. With our parameters, $2\kappa\theta = 0.08 < \sigma_v^2 = 4$, so the condition is not satisfied. This means the variance process can reach zero with positive probability, and Euler-Maruyama discretisation schemes that produce negative variance values would require clamping, introducing bias. Exact CIR sampling, described below, avoids this problem entirely.

The Heston simulator runs on CPU using NumPy and SciPy rather than on GPU. This is because the exact CIR sampling relies on `scipy.stats.ncx2`, which has no GPU implementation, and because the simulation is inherently sequential across time steps: $V_{t+\Delta t}$ depends on V_t, so the $n = 30$ time steps cannot be parallelised. Parallelisation is instead performed across the N paths dimension, with all N non-central chi-squared samples drawn simultaneously at each step. The resulting path tensors are converted to GPU tensors before training and evaluation.

The negative correlation $\rho = -0.70$ reflects the empirical leverage effect, whereby rising volatility is associated with falling asset prices. This feature makes the Heston model qual-

itatively richer than Black & Scholes and more representative of observed equity market dynamics.

3.2.2 Path Generation: Exact CIR Sampling

The variance process is sampled exactly at each time step from its known transition distribution. Conditional on V_t, the next variance $V_{t+\Delta t}$ follows a scaled non-central chi-squared distribution (Broadie and Kaya, 2006):

$$V_{t+\Delta t} \mid V_t \;\sim\; c \cdot \chi^2(df,\; nc(V_t)) \tag{18}$$

with parameters:

$$c = \frac{\sigma_v^2(1 - e^{-\kappa\Delta t})}{4\kappa}, \qquad df = \frac{4\kappa\theta}{\sigma_v^2}, \qquad nc(V_t) = \frac{e^{-\kappa\Delta t}}{c}\, V_t \tag{19}$$

where c is a scale factor, df is the degrees of freedom parameter determined by the ratio of mean-reversion strength to variance noise, and $nc(V_t)$ is the non-centrality parameter that carries the current variance forward. This exact sampling guarantees $V_{t+\Delta t} \geq 0$ for all paths without any clamping or reflection, and incurs no discretisation error in the variance dimension. The parameters c, df, and $e^{-\kappa\Delta t}$ are constant across time steps and are precomputed once before the simulation loop.

3.2.3 Path Generation: Simplified Broadie-Kaya Scheme

Given the sampled variance pair $(V_t, V_{t+\Delta t})$, the integrated variance over the interval is approximated by the trapezoidal rule:

$$I_t = \frac{V_t + V_{t+\Delta t}}{2} \cdot \Delta t \tag{20}$$

The correlated Brownian motions B_t and W_t are decomposed via the Cholesky factorisation as $B_t = \rho W_t + \sqrt{1-\rho^2} W_t^{\perp}$, where $W_t^{\perp}$ is a standard Brownian motion independent of W_t. The component $\int_t^{t+\Delta t} \sqrt{V_s}\, dW_s$ can then be recovered exactly from the sampled pair $(V_t, V_{t+\Delta t})$ by applying the CIR SDE, which gives:

$$V_{t+\Delta t} - V_t = \kappa(\theta - \overline{V})\Delta t + \sigma_v \int_t^{t+\Delta t} \sqrt{V_s}\, dW_s \tag{21}$$

Rearranging and substituting into the log-return formula, the full log-return of the asset price conditional on I_t is:

$$\log \frac{S^1_{t+\Delta t}}{S^1_t} = -\frac{1}{2} I_t + \frac{\rho}{\sigma_v}(V_{t+\Delta t} - V_t - \kappa\theta\,\Delta t + \kappa I_t) + \sqrt{1-\rho^2}\,\sqrt{I_t}\,Z, \quad Z \sim \mathcal{N}(0,1) \tag{22}$$

The first term $-\frac{1}{2}I_t$ is the Itô correction. The second term captures the component of the log-return perfectly correlated with the variance innovation, recovered exactly from the sampled variance pair. The third term is the independent Gaussian component scaled by the residual volatility $\sqrt{1-\rho^2}\,\sqrt{I_t}$. The only approximation in this scheme is the trapezoidal rule for I_t; the log-return given I_t is otherwise exact. This constitutes the simplified Broadie-Kaya scheme of Broadie and Kaya (2006).

3.2.4 Variance Swap as Second Hedging Instrument

Following Buehler et al. (2019) (equations 5.3 and 5.4), the price of a variance swap with maturity T at time t is:

$$S^2_t = \int_0^t V_s\,ds + L(t, V_t) \tag{23}$$

where the function L represents the conditional expectation of future integrated variance given the current state:

$$L(t, v) = \frac{v-\theta}{\kappa}\left(1 - e^{-\kappa(T-t)}\right) + \theta(T-t) \tag{24}$$

The variance swap price therefore decomposes into the realised variance accumulated from inception to t, plus the risk-neutral expectation of the remaining integrated variance given V_t. In our simulator, the running integral $\int_0^t V_s\,ds$ is accumulated step-by-step using the same trapezoidal approximation I_t, so no additional approximation is introduced beyond what is already present in the log-price scheme.

The economic rationale for using the variance swap as a second hedging instrument is that it provides direct and liquid exposure to the variance process V_t, which is the primary driver of option vega risk in stochastic volatility models. Combining stock and variance swap hedging allows the agent to simultaneously hedge both the delta and the vega of the option, which is impossible with the stock alone since V_t is not directly tradable. This two-instrument structure is central to the Heston experiments and is what makes the problem genuinely two-dimensional.

The initial variance swap price is $S^2_0 = L(T, V_0) \approx 0.0033$ with our parameters, giving a normalisation factor of $S_0/L_0 \approx 30{,}380$. This extreme scale disparity between the two instruments makes the variance-swap normalisation described in Section 2 not merely a convenience but a necessity for stable training.

The at-the-money Heston call premium is computed via Fourier inversion at (S_0, V_0, T) as $p_0 = 1.6444$. The paper reports a value of 1.69; the small discrepancy arises from the trapezoidal approximation for the integrated variance in our simplified Broadie-Kaya scheme, which introduces a slight bias in the simulated asset price distribution relative to the exact scheme used by Buehler et al. (2019).

3.2.5 Heston Option Pricer via Fourier Inversion

The Heston call price is computed using the Gil-Pelaez Fourier inversion formula:

$$C(S, V, \tau) = S\,P_1 - Ke^{-r\tau}P_2 \tag{25}$$

where P_1 and P_2 are risk-neutral probabilities recovered from the Heston characteristic function $\varphi(u; S, V, \tau)$ via:

$$P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-iu\log K}\,\varphi_j(u)}{iu}\right] du, \quad j = 1, 2 \tag{26}$$

The Heston characteristic function is available in closed form and is evaluated in double precision (float64) to ensure numerical stability in the complex exponentials. The integral is approximated using 64-point Gauss-Legendre quadrature on the interval $[0, U]$ with $U = 80$, which is sufficient for the parameter range used here. The quadrature nodes and weights are precomputed once at instantiation of the pricer object and reused across all subsequent pricing calls, making repeated evaluation efficient.

3.2.6 Numerical Greeks and Classical Heston Baseline

The stock delta δ_1 and variance-swap delta δ_2 (paper equation 5.6) are obtained by central finite differences applied directly to the Fourier pricer:

$$\delta_1 = \frac{\partial C}{\partial S} \approx \frac{C(S+h,\, V,\, \tau) - C(S-h,\, V,\, \tau)}{2h}, \quad h = 0.01 \cdot S \tag{27}$$

$$\delta_2 = \frac{\partial C/\partial V}{\partial L/\partial V}, \qquad \frac{\partial C}{\partial V} \approx \frac{C(S,\, V+v_b,\, \tau) - C(S,\, V-v_b,\, \tau)}{2v_b}, \quad v_b = 0.001 \tag{28}$$

where the denominator is the analytical sensitivity of the variance swap price to the current variance:

$$\frac{\partial L}{\partial V} = \frac{1 - e^{-\kappa\tau}}{\kappa} \tag{29}$$

The relative bump $h = 0.01 \cdot S$ ensures that the bump size scales appropriately across the distribution of spot prices in the test set, with a minimum absolute bump of 10^{-4} to avoid numerical issues at very small spot values. At expiry ($\tau = 0$), the stock delta degenerates

to $\mathbf{1}_{S>K}$, which is handled explicitly. The denominator $\partial L/\partial V$ is clamped below by 10^{-8} to prevent division by zero near expiry.

The Heston baseline hedge is constructed by computing (δ_1, δ_2) at each of the $n = 30$ rebalancing dates across all N test paths and applying the terminal P&L formula of equation (1) with the variance-swap normalisation described in Section 2. Because the Fourier pricer must be called twice per greek, two greeks, 30 time steps and 50,000 paths, the baseline requires $2 \times 2 \times 30 \times 50{,}000 = 6{,}000{,}000$ pricer evaluations, each involving 64-point quadrature. This makes the baseline computation substantially more expensive than training the deep hedge: evaluating the Heston baseline on the full test set requires 114.5 seconds on CPU, compared to roughly 62 seconds to train the deep hedge on GPU for 1500 epochs. This confirms the scalability claim of Buehler et al. (2019): the deep hedge is not only more flexible under market frictions but also computationally competitive with the classical approach.

4 Numerical Experiments

This section presents the numerical experiments conducted in two market environments. The Black & Scholes setting serves as a validation environment where the analytical delta provides a theoretically optimal benchmark, allowing us to verify that the deep hedging pipeline is correctly implemented before studying more complex dynamics. The Heston stochastic volatility setting is the primary environment of interest, where the presence of unhedgeable volatility risk and the use of a variance swap as a second instrument make the problem genuinely two-dimensional. Within each environment we study four aspects: the frictionless case as a correctness check, the effect of proportional transaction costs as the central experiment, the comparison between CVaR and entropic risk objectives, and a hyperparameter sensitivity analysis covering network width and batch size. The section concludes with the numerical verification of the $O(\epsilon^{2/3})$ price asymptotics for the exponential utility indifference price.

4.1 Black & Scholes

Implementation Choices

All Black & Scholes experiments use the `per_step` policy, that is one dedicated network per rebalancing date, with batch normalisation disabled. Although the original paper applies batch normalisation, we found empirically that it destabilised training under on-policy simulation: enabling it increased out-of-sample CVaR(95%) relative to the disabled setting, and we therefore disabled it throughout. We use 500 training epochs, which is sufficient for

convergence in the Black & Scholes setting as confirmed by the training curves in Figure 3a, and a batch size of 4096, which yields more stable gradient estimates than the paper's default of 256 as demonstrated in the sensitivity analysis of Section 4.1.3. The specific GPU model used for all Black & Scholes experiments is an NVIDIA RTX 5070 Ti laptop GPU, with training times of approximately 10 to 13 seconds per model.

4.1.1 Black & Scholes Without Transaction Costs

The analytical delta hedge achieves a CVaR(95%) of losses of 0.8221 in the frictionless setting, reflecting the discretisation error introduced by daily rebalancing over 30 steps. In continuous time this residual would vanish entirely; its small magnitude confirms that the path simulator and baseline implementation are correct before introducing market frictions.

Results

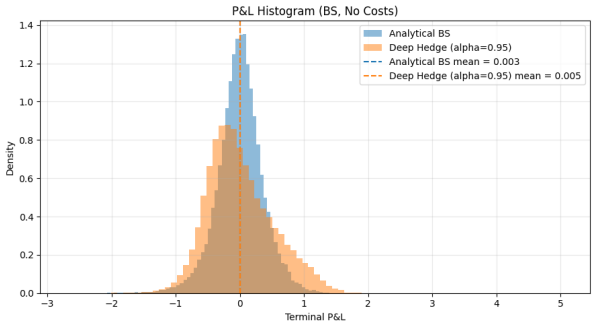
Table 3: Black & Scholes, no transaction costs ($\epsilon = 0$): out-of-sample risk metrics. All runs on NVIDIA RTX 5070 Ti (laptop), 500 epochs, batch size 4096.

Metric	**BS Baseline**	**DH** $\alpha = 0.50$	**DH** $\alpha = 0.95$	**DH** $\alpha = 0.99$
Mean P&L	0.0032	0.0042	0.0046	0.0052
Std P&L	0.3548	0.4715	0.5180	0.5031
VaR (95%)	0.5805	0.7593	0.7288	0.6890
CVaR (95%)	0.8221	1.0700	0.9219	0.8589
VaR (99%)	0.9667	1.2571	1.0289	0.9607
CVaR (99%)	1.2187	1.5919	1.2342	1.1284
Turnover	2.2238	1.8849	1.8604	1.8676
Avg Transaction Cost	0.0000	0.0000	0.0000	0.0000
Training Time (s)	—	11.2	11.4	11.4

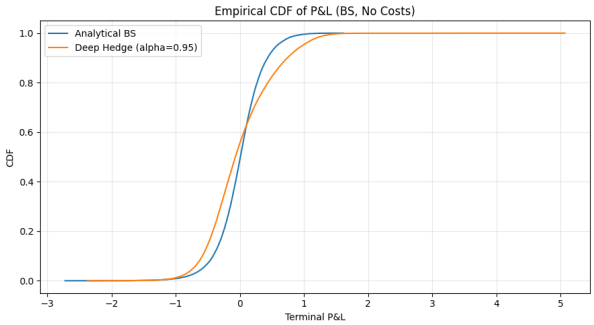
Table 3 reports the out-of-sample risk metrics for the Black & Scholes frictionless setting ($\epsilon = 0$). As expected, the analytical delta hedge is difficult to improve upon: it is the theoretically optimal variance-minimising strategy in a complete frictionless market, and any residual risk arises purely from the discrete rebalancing schedule rather than from a suboptimal hedging policy. The deep hedge with $\alpha = 0.5$ performs slightly worse on all tail metrics. This is consistent with the gradient signal from a median-focused CVaR objective covering only the worst 50% of paths, which provides insufficient pressure to improve upon the analytical delta in a setting where it is already the variance-minimising strategy. At $\alpha = 0.99$, the deep hedge achieves a marginal improvement in CVaR(99%)

(1.1284 vs. 1.2187), consistent with the observation of Buehler et al. (2019) that higher risk-aversion levels shift the learned strategy toward protecting the extreme tail at the cost of higher overall variance. Turnover is slightly lower for all deep hedge variants relative to the analytical baseline, suggesting the network learns to rebalance less frequently even in the absence of explicit cost penalisation. This experiment primarily serves as a correctness check: the deep hedge reproduces behaviour close to the analytical benchmark, validating both the simulator and the training pipeline before introducing market frictions.

Figures

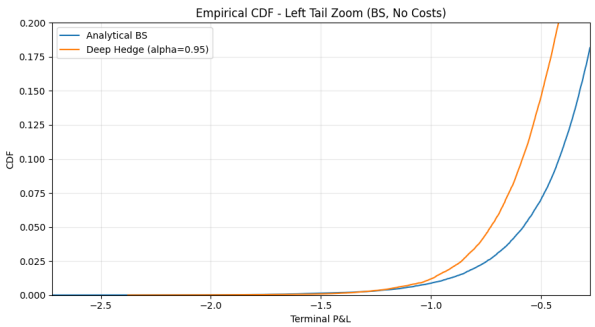


(a) P&L histogram: Analytical BS vs. Deep Hedge ($\alpha = 0.95$).

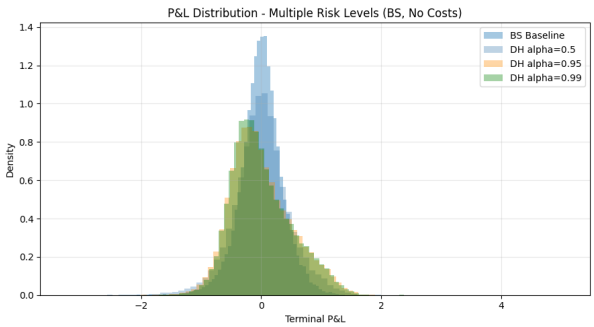


(b) Empirical CDF of terminal P&L.

Figure 1: P&L distribution comparison, Black & Scholes without transaction costs ($\epsilon = 0$), $\alpha = 0.95$.

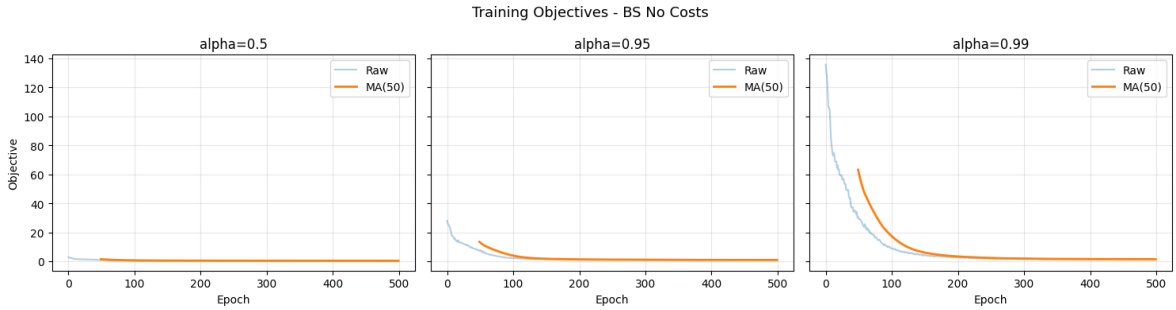


(a) Left-tail zoom of empirical CDF (bottom 20%).

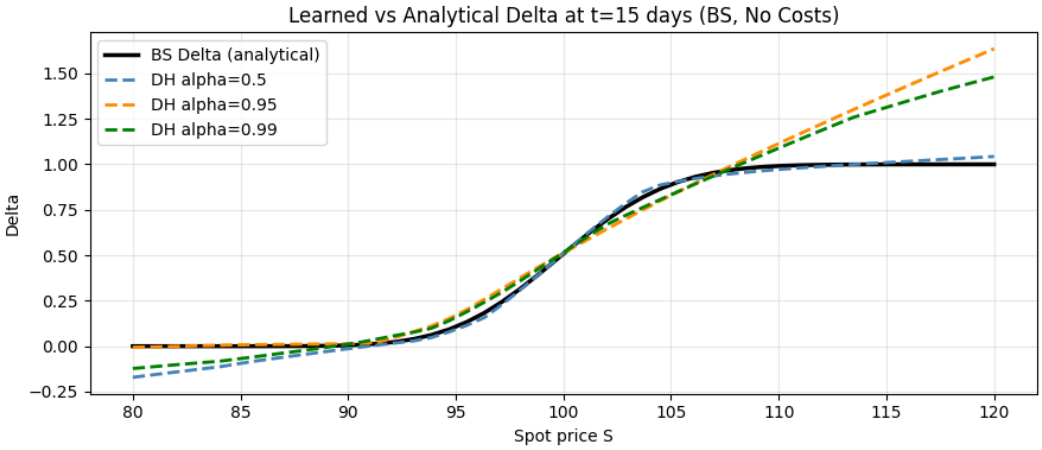


(b) P&L distributions across all risk levels.

Figure 2: Tail analysis and multi-level comparison, Black & Scholes without transaction costs ($\epsilon = 0$).



(a) Training objectives (raw and MA(50)) for all α levels.



(b) Learned vs. analytical delta at $t = 15$ days.

Figure 3: Training diagnostics and delta comparison, Black & Scholes without transaction costs ($\epsilon = 0$).

The P&L histogram (Figure 1a) confirms that the analytical delta produces a tighter distribution centred around zero, while the deep hedge exhibits higher variance, consistent with the metrics in Table 3. The empirical CDF (Figure 1b) and left-tail zoom (Figure 2a) show that in the absence of transaction costs the analytical delta dominates throughout the loss region, as expected from theory. The multi-level histogram (Figure 2b) illustrates that higher α values produce progressively wider distributions, reflecting the trade-off between tail protection and overall variance: the $\alpha = 0.99$ strategy concentrates gradient updates on the extreme left tail and therefore accepts higher variance across the bulk of the distribution in exchange for reduced extreme losses.

Training curves (Figure 3a) confirm clean convergence for all three risk levels within 500 epochs, with the bulk of optimisation occurring in the first 100 epochs and a stable plateau thereafter. The initial objective value scales with α, reaching approximately 140 for $\alpha = 0.99$, reflecting the greater weight assigned to extreme tail losses by the CVaR objective at high confidence levels.

The delta comparison (Figure 3b) shows that near the money ($S \in [95, 110]$) all three deep hedge variants closely track the analytical Black & Scholes delta, validating the network's

ability to recover the optimal frictionless strategy in the region most relevant for hedging. At extreme spot values ($S < 90$ and $S > 110$), the learned deltas deviate from the analytical benchmark. This is not a failure of generalisation per se but rather a consequence of the CVaR objective: deep out-of-the-money and deep in-the-money paths contribute negligibly to the tail loss metric, providing insufficient gradient pressure to enforce accurate delta recovery in these regions. The resulting deviations have little practical impact on hedging performance since these regions are rarely visited by the simulated paths.

4.1.2 Black & Scholes With Transaction Costs

With proportional transaction costs $\epsilon = 0.005$, the analytical delta hedge incurs an average cost of 1.1116 per path, degrading its CVaR(95%) to 2.3970 compared to 0.8221 in the frictionless case. This deterioration is structural: the analytical delta is computed without any awareness of transaction costs, so it rebalances at full intensity at every step regardless of the friction incurred. The deep hedge, by contrast, is trained with the cost term explicitly included in the P&L objective and learns to trade less aggressively, achieving substantially better tail risk metrics across all confidence levels. Note that the DH model is evaluated at $\epsilon = 0.005$, its training cost level, and separately scanned across a broader range of cost levels in Figure 7.

Table 4: Black & Scholes, with transaction costs ($\epsilon = 0.005$): out-of-sample risk metrics. All runs on NVIDIA RTX 5070 Ti (laptop), 500 epochs, batch size 4096.

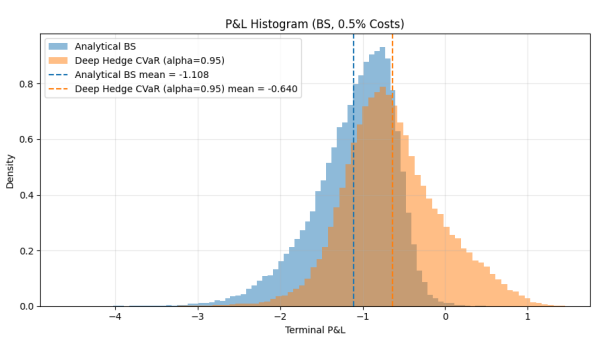
Metric	**BS Baseline**	**DH** $\alpha = 0.50$	**DH** $\alpha = 0.95$	**DH** $\alpha = 0.99$
Mean P&L	-1.1084	-0.5333	-0.6401	-0.6303
Std P&L	0.5033	0.7034	0.5910	0.6120
VaR (95%)	2.0594	1.7667	1.5189	1.5307
CVaR (95%)	2.3970	2.2369	1.8206	1.8459
VaR (99%)	2.6005	2.5157	2.0014	2.0263
CVaR (99%)	2.9221	2.9779	2.3187	2.3682
Turnover	2.2238	1.0806	1.2918	1.2705
Avg Transaction Cost	1.1116	0.5404	0.6459	0.6352
Training Time (s)	—	11.3	11.3	10.8

Table 4 demonstrates that the deep hedge provides decisive benefits once transaction costs are present, confirming the central claim of Buehler et al. (2019). At $\alpha = 0.95$, CVaR(95%) is reduced by 24% (from 2.3970 to 1.8206) and CVaR(99%) by 21% (from 2.9221 to 2.3187). Turnover falls from 2.22 to 1.29 and average transaction costs from 1.1116 to 0.6459, a saving of 42%. The less negative mean P&L (-0.64 vs. -1.11) confirms that the network retains

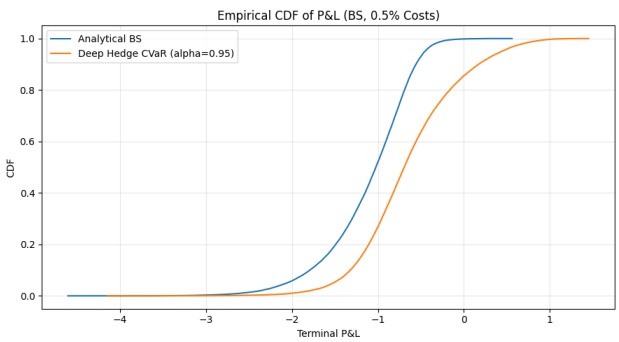
significantly more of the received premium by trading less aggressively. These improvements arise entirely from the network learning to internalise the cost of rebalancing, which is a capability the analytical delta hedge lacks by construction.

At $\alpha = 0.5$ the improvements are more modest and CVaR(99%) is marginally worse (2.9779 vs. 2.9221). This is consistent with the sparse gradient signal of the median-focused CVaR objective: with only 50% of paths contributing to the gradient, the penalty on extreme tail losses is insufficient to fully overcome the cost-saving incentive of reducing turnover. The $\alpha = 0.95$ and $\alpha = 0.99$ strategies yield nearly identical results, suggesting that both converge to a similar cost-aware rebalancing policy. Training remains fast at approximately 11 seconds per model on the NVIDIA RTX 5070 Ti laptop GPU.

Figures

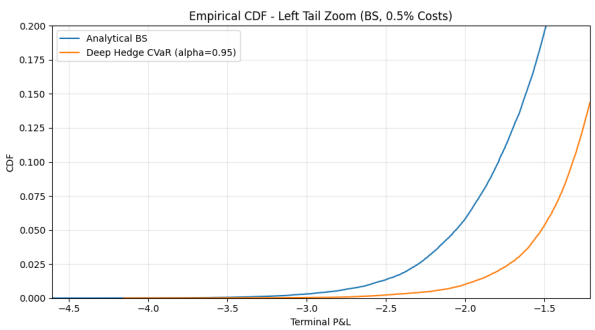


(a) P&L histogram: Analytical BS vs. Deep Hedge ($\alpha = 0.95$).

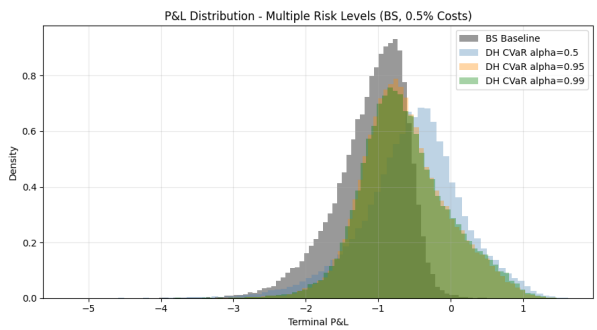


(b) Empirical CDF of terminal P&L.

Figure 4: P&L distribution comparison, Black & Scholes with transaction costs ($\epsilon = 0.005$), $\alpha = 0.95$.

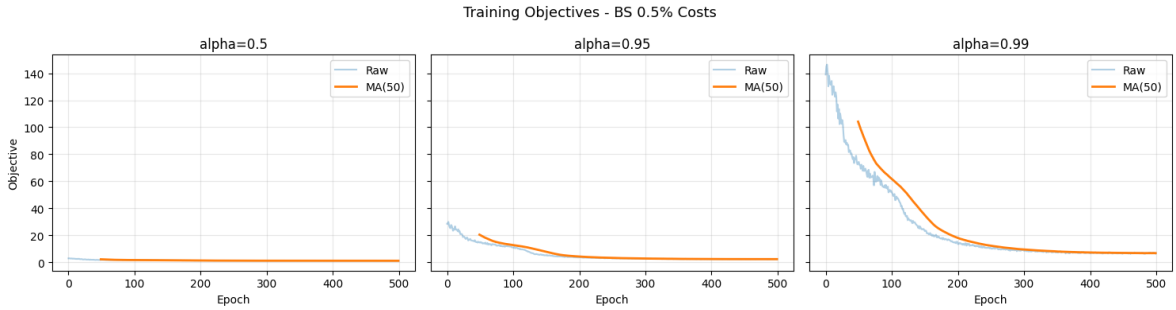


(a) Left-tail zoom of empirical CDF (bottom 20%).

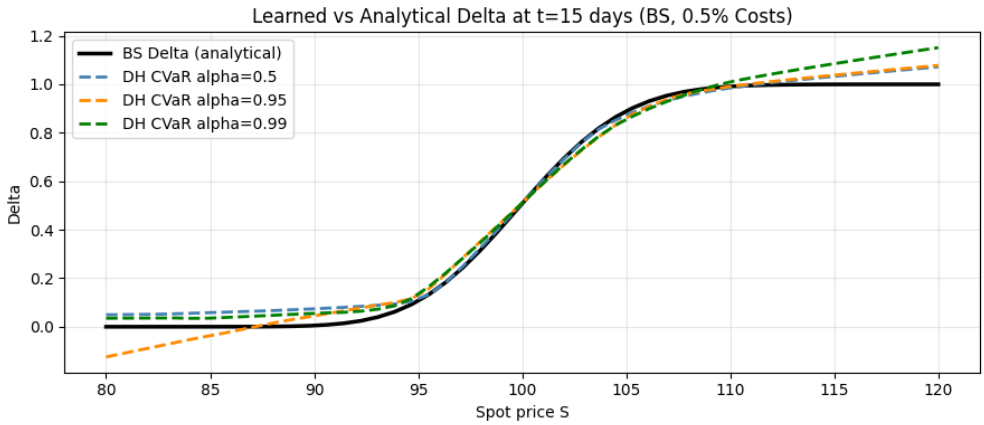


(b) P&L distributions across all risk levels.

Figure 5: Tail analysis and multi-level comparison, Black & Scholes with transaction costs ($\epsilon = 0.005$).



(a) Training objectives (raw and MA(50)) for all α levels.



(b) Learned vs. analytical delta at $t = 15$ days.

Figure 6: Training diagnostics and delta comparison, Black & Scholes with transaction costs ($\epsilon = 0.005$).

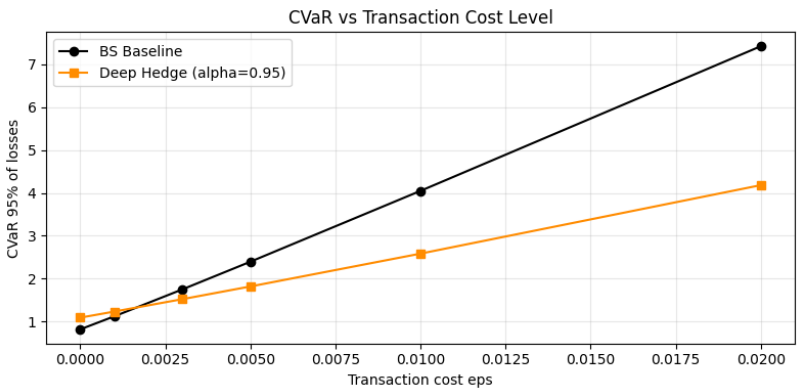


Figure 7: CVaR(95%) as a function of transaction cost level ϵ: Analytical BS baseline vs. Deep Hedge ($\alpha = 0.95$), Black & Scholes. The DH model is fixed at the policy trained at $\epsilon = 0.005$ and is not retrained at each cost level.

The P&L histogram (Figure 4a) illustrates the central result of this experiment: the deep hedge distribution is shifted substantially to the right of the analytical baseline, with a mean P&L of -0.64 compared to -1.11, and its left tail truncated well before that of the baseline. The full empirical CDF (Figure 4b) shows that the deep hedge stochastically dominates the analytical delta across the entire distribution once transaction costs are present.

The left-tail zoom (Figure 5a) makes this particularly vivid: the analytical delta accumulates probability mass in the extreme loss region far more rapidly, confirming the CVaR improvements reported in Table 4.

The multi-level histogram (Figure 5b) shows that all three deep hedge variants outperform the baseline in the loss region, with the $\alpha = 0.95$ and $\alpha = 0.99$ strategies nearly indistinguishable, consistent with both converging to a similar cost-aware rebalancing policy. The $\alpha = 0.5$ strategy is visibly shifted less far to the right, reflecting its weaker tail penalty.

Training curves (Figure 6a) confirm clean convergence within 500 epochs for all risk levels. The delta comparison (Figure 6b) shows that near the money all learned strategies closely track the analytical delta. Notably, deep out-of-the-money deltas ($S < 90$) remain slightly above zero rather than collapsing to the analytical near-zero value: the network learns to avoid full liquidation in order to save on transaction costs, an economically sensible adaptation that has no analogue in the cost-oblivious analytical hedge.

Figure 7 presents CVaR(95%) as a function of ϵ for both strategies. At $\epsilon = 0$ the analytical delta achieves lower CVaR, as expected from the frictionless theory. As ϵ increases beyond approximately 0.002, the deep hedge consistently dominates and the gap widens rapidly: at $\epsilon = 0.02$ the baseline CVaR reaches 7.4 while the deep hedge remains at 4.2. This confirms that the benefit of cost-aware hedging is not limited to the training cost level of 0.5% but generalises across a broad range of market frictions. Since the DH model is evaluated at its fixed trained policy rather than retrained at each ϵ, a model retrained specifically at each cost level would show an even stronger advantage.

4.1.3 Hyperparameter Sensitivity: Black & Scholes

Table 5: Network width sensitivity: CVaR metrics, turnover and training time at $\alpha = 0.95$, $\epsilon = 0.005$. All runs on NVIDIA RTX 5070 Ti (laptop), 500 epochs, batch size 4096.

Config	**Mean P&L**	**CVaR(95%)**	**CVaR(99%)**	**Turnover**	**Time (s)**
Small $[8, 8]$	-0.5787	2.1259	2.7583	1.1697	11.1
Paper $[16, 16]$	-0.6211	1.8538	2.3885	1.2531	10.9
Large $[32, 32]$	-0.6381	1.8315	2.3764	1.2868	11.0

Table 6: Batch size sensitivity: CVaR metrics, turnover and training time at $\alpha = 0.95$, $\epsilon = 0.005$, hidden dims $[16, 16]$. All runs on NVIDIA RTX 5070 Ti (laptop), 500 epochs.

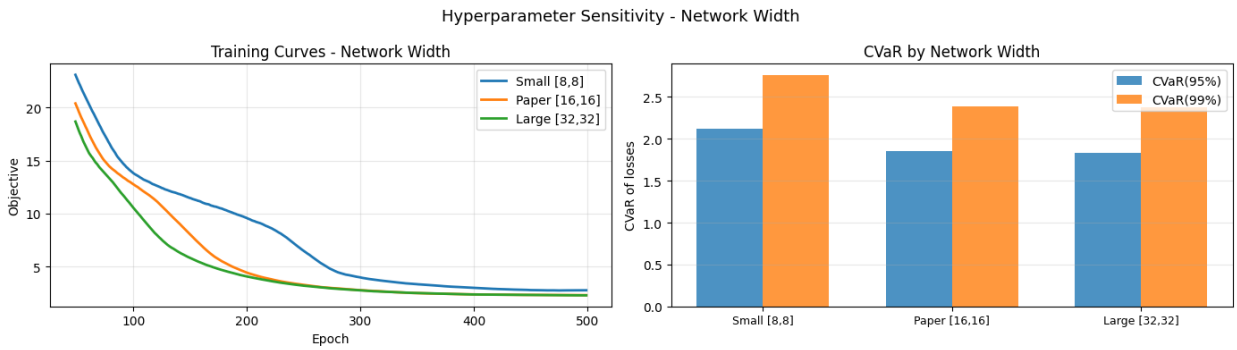
Config	**Mean P&L**	**CVaR(95%)**	**CVaR(99%)**	**Turnover**	**Time (s)**
Batch 64	-0.5366	2.4185	3.1770	1.0885	10.8
Batch 256	-0.5913	1.9704	2.5624	1.1949	10.7
Batch 512	-0.5881	1.9642	2.5364	1.1911	10.7
Batch 4096	-0.6342	1.8324	2.3546	1.2784	10.5

Table 5 shows that the paper's architecture $[16, 16]$ meaningfully outperforms the smaller $[8, 8]$ network (CVaR(95%) 1.85 vs. 2.13), confirming that the network requires sufficient capacity to represent the cost-aware hedging policy. Increasing to $[32, 32]$ yields only marginal further improvement (1.83 vs. 1.85), indicating clear diminishing returns beyond the paper's design. The training curves in Figure 8 show that all three widths converge at a similar rate, with no sign of underfitting for $[16, 16]$.

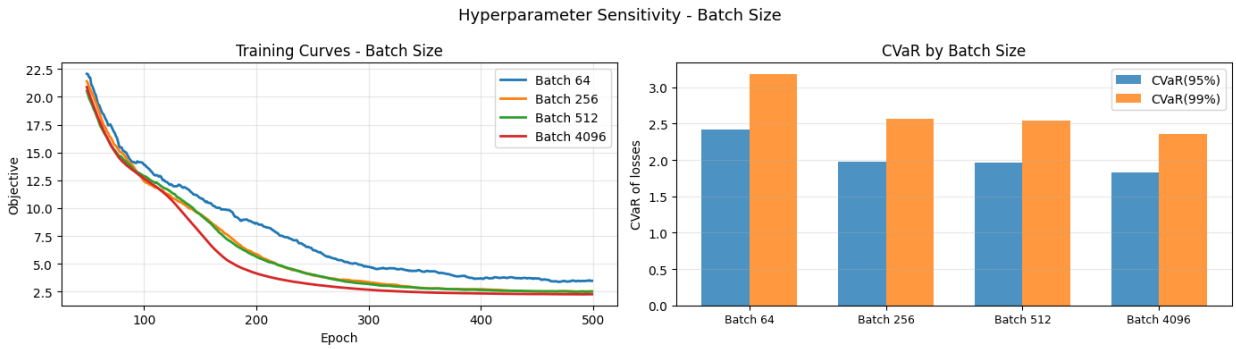
Table 6 reveals a broadly monotone improvement in CVaR from batch size 64 to 4096, driven by increasingly stable gradient estimates. The improvement from batch 512 to 4096 is marginal (CVaR(95%) 1.96 to 1.83), suggesting that 512 is already near the point of diminishing returns for the Black & Scholes setting. Batch 64 is clearly suboptimal, exhibiting noisier training dynamics as visible in Figure 8. We use batch size 4096 throughout as it achieves the best tail metrics with negligible additional training time relative to smaller batches.

Training time remains stable at approximately 10 to 11 seconds across all width and batch configurations, confirming that the computational cost of the deep hedge is largely invariant to these hyperparameter choices in the Black & Scholes setting. This is consistent with the scalability claim of Buehler et al. (2019): the per-epoch cost is dominated by path simulation and the forward pass rather than by network size in this one-dimensional environment.

Figures



(a) Network width sensitivity: training curves (left) and CVaR bar chart (right)



(b) Batch size sensitivity: training curves (left) and CVaR bar chart (right)

Figure 8: Hyperparameter sensitivity analysis, Black & Scholes with transaction costs ($\epsilon = 0.005$, $\alpha = 0.95$)

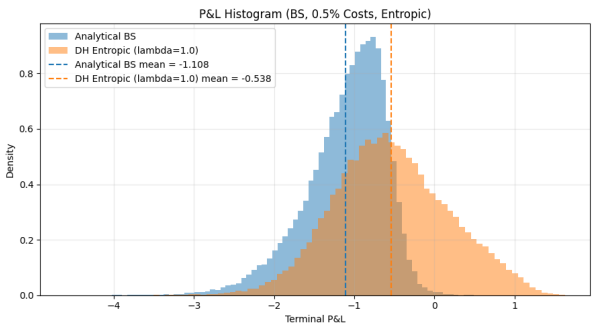
4.1.4 Black & Scholes: Entropic Risk Measure

Table 7: Black & Scholes, $\epsilon = 0.005$: three-way comparison between the analytical baseline, Deep Hedge CVaR ($\alpha = 0.95$) and Deep Hedge Entropic ($\lambda = 1$). Entropic model trained in 11.1s on NVIDIA RTX 5070 Ti (laptop), 500 epochs, batch size 4096.

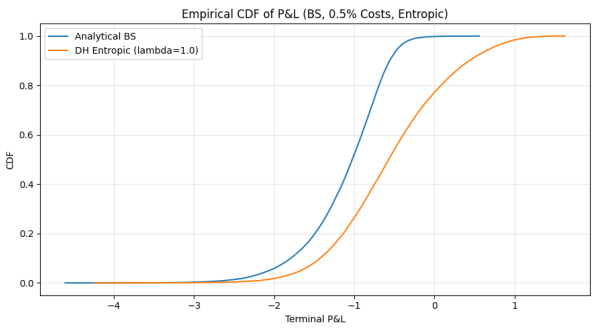
Metric	**BS Baseline**	**DH CVaR** $\alpha = 0.95$	**DH Entropic** $\lambda = 1$
Mean P&L	-1.1084	-0.6401	-0.5377
Std P&L	0.5033	0.5910	0.7179
VaR (95%)	2.0594	1.5189	1.6671
CVaR (95%)	2.3970	1.8206	1.9899
VaR (99%)	2.6005	2.0014	2.1746
CVaR (99%)	2.9221	2.3187	2.5028
Turnover	2.2238	1.2918	1.0908
Avg Transaction Cost	1.1116	0.6459	0.5459

Table 7 compares the two deep hedge objectives against the analytical baseline. Both outperform the baseline on all tail metrics, confirming that the benefit of cost-aware hedging is robust to the choice of risk measure. The entropic model ($\lambda = 1$) achieves the best mean P&L (-0.54 vs. -0.64) and the lowest turnover (1.09 vs. 1.29), trading the least aggressively of all three strategies and saving 51% in average transaction costs relative to the analytical baseline. However, its tail metrics are worse than those of the CVaR model: CVaR(95%) is 1.99 compared to 1.82. This is the expected trade-off: the entropic objective penalises deviations from the mean symmetrically through the exponential utility function, without specifically targeting the tail. The CVaR objective, by construction, focuses optimisation effort on the worst $(1 - \alpha)$ outcomes and therefore achieves superior tail protection at the cost of higher average turnover. Both objectives are valid choices depending on whether the practitioner prioritises cost minimisation or tail risk reduction, and neither dominates the other across all metrics.

Figures

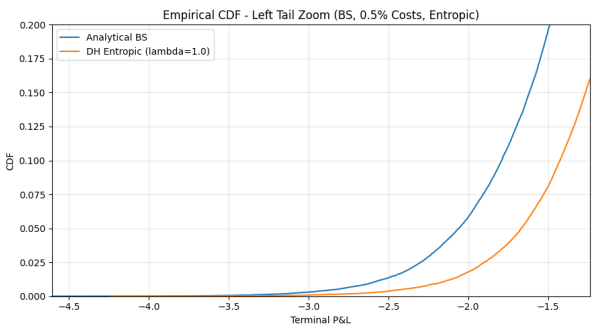


(a) P&L histogram: Analytical BS vs. Deep Hedge Entropic ($\lambda = 1$).

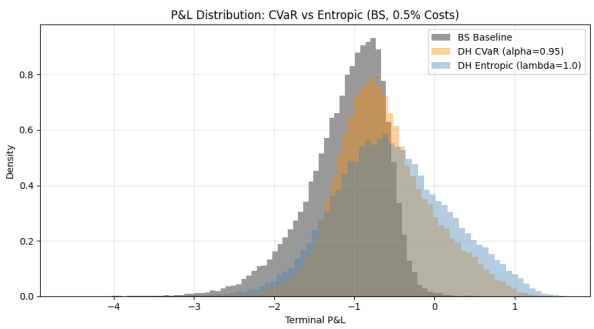


(b) Empirical CDF of terminal P&L.

Figure 9: P&L distribution comparison, Black & Scholes with transaction costs ($\epsilon = 0.005$), Entropic objective ($\lambda = 1$).

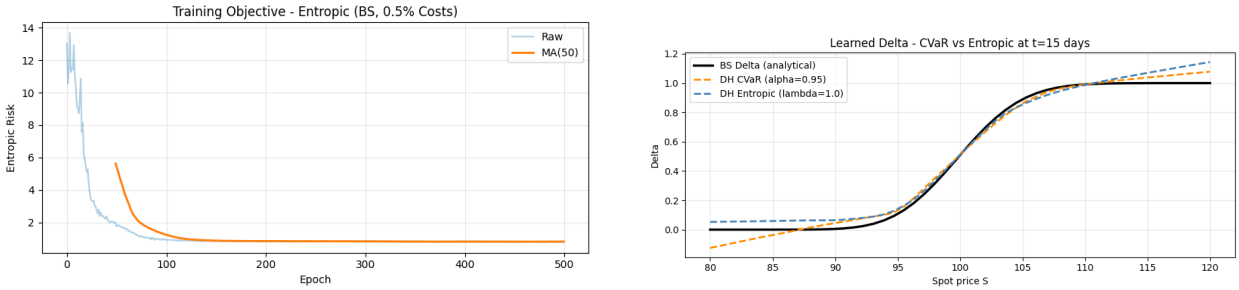


(a) Left-tail zoom of empirical CDF (bottom 20%).



(b) P&L distributions: baseline vs. CVaR vs. Entropic.

Figure 10: Tail analysis and three-way comparison, Black & Scholes with transaction costs ($\epsilon = 0.005$).



(a) Training objective (Entropic risk, raw and MA(50)).

(b) Learned delta: CVaR vs. Entropic at $t = 15$ days.

Figure 11: Training diagnostics and delta comparison, Black & Scholes Entropic objective ($\epsilon = 0.005$).

The histogram (Figure 9a) and full CDF (Figure 9b) confirm that the entropic model substantially outperforms the analytical baseline, with the distribution shifted right and the left tail truncated. The left-tail zoom (Figure 10a) shows that the separation from the baseline is somewhat less sharp than in the CVaR case, consistent with the entropic objective not explicitly targeting tail outcomes. The three-way overlay (Figure 10b) makes the difference between the two deep hedge objectives directly visible: the CVaR distribution is tighter in the loss region while the entropic distribution is slightly wider but shifted further right, reflecting its focus on mean performance over tail protection.

The training curve (Figure 11a) shows rapid convergence within the first 100 epochs and a stable plateau thereafter, consistent with the CVaR training dynamics. The delta comparison (Figure 11b) reveals a qualitative difference between the two objectives in the deep out-of-the-money region ($S < 90$): the entropic model maintains a small positive delta while the CVaR model goes slightly negative. This reflects the different risk geometries of the two objectives: the entropic penalty is symmetric around the mean, leading to a more conservative position in the tails, whereas the CVaR model accepts small negative deltas to better protect the worst-case outcomes it directly optimises.

4.2 Heston

Implementation Choices

All Heston experiments use the `per_step` policy with batch normalisation disabled, for the same reasons described in the Black & Scholes implementation choices. Compared to Black & Scholes, the Heston setting requires 1500 training epochs to reach stable convergence, reflecting the increased complexity of two-asset hedging under stochastic volatility. Although the training curves suggest apparent convergence within the first 200 to 300 epochs, we found empirically that continuing to train until 1500 epochs yields meaningfully better tail

metrics: models stopped early exhibit higher CVaR values on the test set, suggesting that the network continues to refine its hedging policy in the later epochs despite the objective appearing flat. This is consistent with the known behaviour of Adam with cosine annealing, where the learning rate decay in later epochs allows fine-grained parameter adjustments that do not visibly move the training loss but improve out-of-sample generalisation. Batch size is set to 4096 and hidden dimensions to $[17, 17]$, matching the paper's prescription of $N_1 = N_2 = d + 15$ with $d = 2$. The Heston call premium is computed via Fourier inversion at (S_0, V_0, T) as $p_0 = 1.6444$, slightly below the paper's reported value of 1.69, with the small discrepancy attributable to the trapezoidal approximation for integrated variance in the simplified Broadie-Kaya scheme as discussed in Section 3. Training times are approximately 65 to 70 seconds per model on the NVIDIA RTX 5070 Ti laptop GPU, compared to 114.5 seconds required to evaluate the numerical Heston baseline greeks across the full test set.

4.2.1 Heston Without Transaction Costs

Table 8: Heston, no transaction costs ($\epsilon = 0$): out-of-sample risk metrics. Model trained in 65.5s on NVIDIA RTX 5070 Ti (laptop), 1500 epochs, batch size 4096.

Metric	**Model Delta**	**DH CVaR** $\alpha = 0.5$
Mean P&L	-0.0440	-0.0442
Std P&L	0.4307	0.6479
VaR (95%)	0.6716	1.1692
CVaR (95%)	1.0405	1.8125
VaR (99%)	1.2494	2.1878
CVaR (99%)	1.7180	2.9019
Turnover	2.2564	3.6937
Turnover d1 (stock units)	2.1790	3.5475
Turnover d2 (var-swap/S_0)	0.0774	0.1463
Avg Transaction Cost	0.0000	0.0000

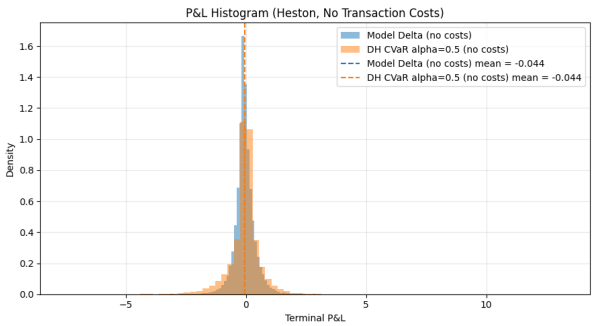
As in the Black & Scholes frictionless case, this experiment serves primarily as a correctness and calibration check. The Heston model delta hedge computes both the stock delta $\delta_1 = \partial C/\partial S$ and the variance-swap delta $\delta_2 = (\partial C/\partial V)/(\partial L/\partial V)$ (paper eq. 5.6) via central finite differences on the Fourier pricer at each rebalancing date, providing a near-optimal benchmark in the frictionless two-asset setting.

Table 8 shows that the model delta achieves substantially lower CVaR than the deep hedge (1.04 vs. 1.81 at the 95% level), consistent with it being the theoretically motivated bench-

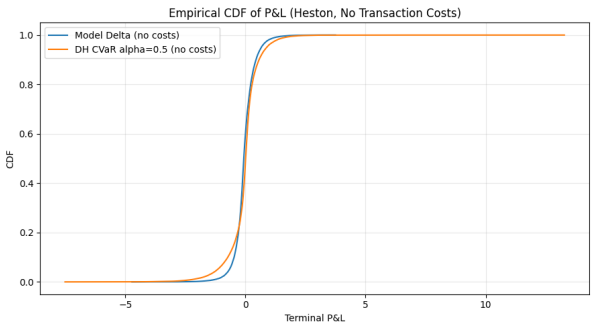
mark in the absence of frictions. Both strategies share essentially identical mean P&L (-0.044), confirming that neither strategy introduces bias. The deep hedge exhibits significantly higher variance (0.65 vs. 0.43), which is expected: with $\alpha = 0.5$ the CVaR objective approximates a mean-loss criterion, so the network optimises something close to variance minimisation. In the frictionless two-asset Heston setting, the Fourier-computed model delta is already the theoretically motivated variance-minimising strategy, leaving the network no structural advantage to exploit while introducing approximation noise from the finite-capacity neural parametrisation.

The turnover decomposition reveals that hedging activity is almost entirely driven by the stock leg: d1 turnover is 2.18 for the model delta and 3.55 for the deep hedge, while d2 turnover in variance-swap units normalised by S_0 is negligible at 0.08 and 0.15 respectively. This confirms that the variance-swap position changes slowly relative to the stock delta, consistent with the smooth and slowly varying behaviour of $L(t, V_t)$ over the 30-day hedging horizon.

Figures



(a) P&L histogram: Model Delta vs. Deep Hedge ($\alpha = 0.5$).



(b) Empirical CDF of terminal P&L.

Figure 12: P&L distribution comparison, Heston without transaction costs ($\epsilon = 0$), $\alpha = 0.5$.

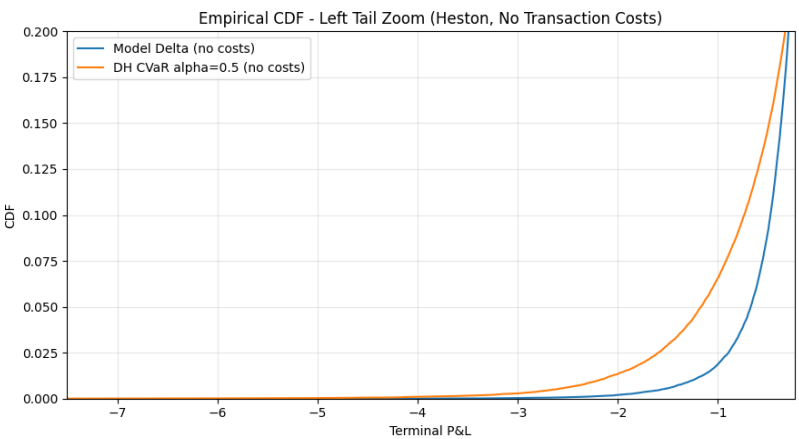


Figure 13: Left-tail zoom of empirical CDF, Heston without transaction costs ($\epsilon = 0$).

The histogram (Figure 12a) shows that both strategies are centred at the same mean, but the deep hedge distribution is considerably wider, with a notable fat right tail extending beyond +10. This arises from paths where the network under-hedges and the option expires worthless, generating large positive P&L. The empirical CDF (Figure 12b) confirms that the two strategies track closely in the loss region but diverge substantially in the right tail. The left-tail zoom (Figure 13) shows that the model delta accumulates slightly more moderate losses in the $[-2, -0.5]$ range, while the deep hedge has marginally heavier extreme losses below -2, consistent with the higher CVaR reported in Table 8. As with the Black & Scholes frictionless experiment, this confirms the correctness of the simulator and baseline before introducing transaction costs.

4.2.2 Heston With Transaction Costs

The numerical Heston baseline requires computing both greeks (δ_1, δ_2) via central finite differences on the Fourier pricer at every rebalancing date across all test paths, taking 114.5 seconds. It achieves a baseline CVaR(95%) of 2.9688 at $\epsilon = 0.005$, substantially higher than the frictionless case (1.0405), confirming that transaction costs significantly degrade the model delta performance in the stochastic volatility setting.

Table 9: Heston, with transaction costs ($\epsilon = 0.005$): out-of-sample risk metrics and training times. All runs on NVIDIA RTX 5070 Ti (laptop), 1500 epochs, batch size 4096.

Metric	**Model Delta**	**DH** $\alpha = 0.50$	**DH** $\alpha = 0.95$	**DH** $\alpha = 0.99$
Mean P&L	-1.1703	-0.2225	-0.4556	-0.4490
Std P&L	0.6463	1.4534	1.4180	1.4636
VaR (95%)	2.4216	2.6838	1.9153	2.0149
CVaR (95%)	2.9688	4.7098	2.3763	2.4140
VaR (99%)	3.2842	5.8640	2.6302	2.6080
CVaR (99%)	3.8624	8.3159	3.3057	3.1998
Turnover	2.2579	0.3463	0.8537	0.8362
Turnover d1 (stock units)	2.1801	0.2837	0.7388	0.7244
Turnover d2 (var-swap/S_0)	0.0778	0.0626	0.1149	0.1118
Avg Transaction Cost	1.1243	0.1723	0.4231	0.4145
Training Time (s)	114.5	67.2	69.8	70.2

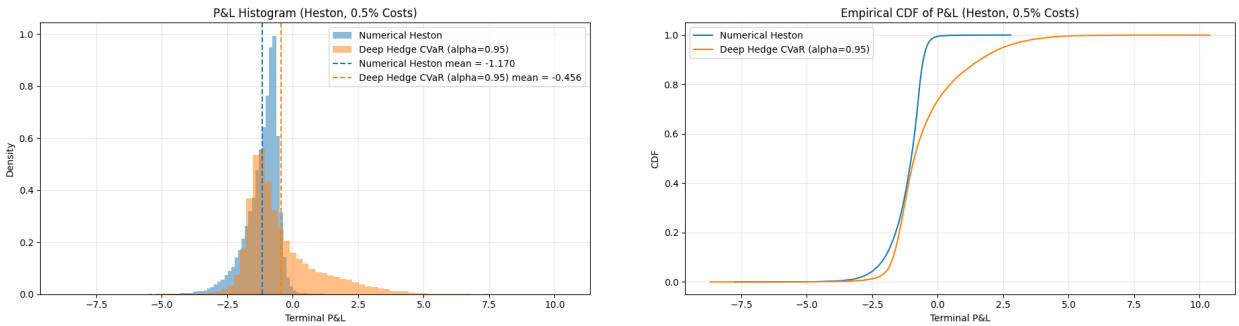
Table 9 reveals an important distinction between the three risk levels. At $\alpha = 0.5$, the deep hedge exhibits a degenerate behaviour: turnover collapses to 0.35, the network has learned to barely hedge, retaining most of the premium and achieving the best mean P&L (-0.22) but at the cost of catastrophic tail risk. CVaR(95%) rises to 4.71, far worse than the model

delta baseline of 2.97. This is a known failure mode of mild risk objectives in complex multi-asset settings: with $\alpha = 0.5$ the CVaR penalty on tail losses is insufficiently strong to prevent the network from discovering the degenerate solution of under-trading to save transaction costs, as the gradient signal covers only the worst 50% of paths and provides insufficient pressure to enforce meaningful hedging behaviour.

At $\alpha = 0.95$ and $\alpha = 0.99$, the deep hedge delivers the expected benefits. CVaR(95%) is reduced by 20% (from 2.9688 to 2.3763) and CVaR(99%) by 14% (from 3.8624 to 3.3057) at $\alpha = 0.95$. Turnover falls from 2.26 to 0.85 and average transaction costs are reduced from 1.1243 to 0.4231, a saving of 62%. The $\alpha = 0.99$ strategy yields nearly identical results, confirming convergence to a similar cost-aware policy. Both tail-focused strategies substantially outperform the model delta baseline, replicating the central finding of Buehler et al. (2019) in the stochastic volatility setting.

The turnover decomposition confirms that hedging activity remains dominated by the stock leg at $\alpha = 0.95$: d1 turnover falls from 2.18 to 0.74 while d2 turnover in variance-swap units normalised by S_0 actually increases slightly from 0.08 to 0.11, suggesting the network learns to hedge variance risk more actively than the model delta when transaction costs are present. Training time is stable at approximately 67 to 70 seconds per model, substantially faster than the 114.5 seconds required to compute the analytical Heston baseline greeks across the full test set.

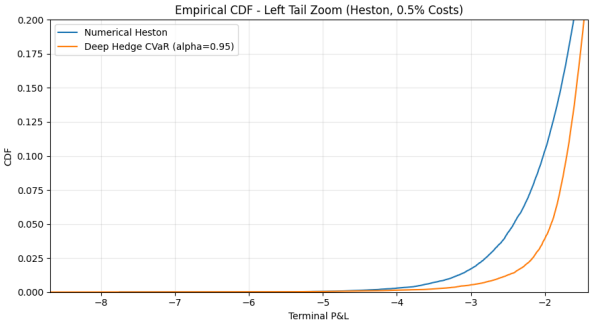
Figures



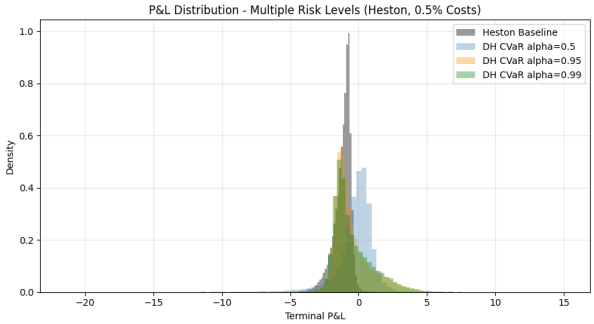
(a) P&L histogram: Model Delta vs. Deep Hedge ($\alpha = 0.95$).

(b) Empirical CDF of terminal P&L.

Figure 14: P&L distribution comparison, Heston with transaction costs ($\epsilon = 0.005$), $\alpha = 0.95$.

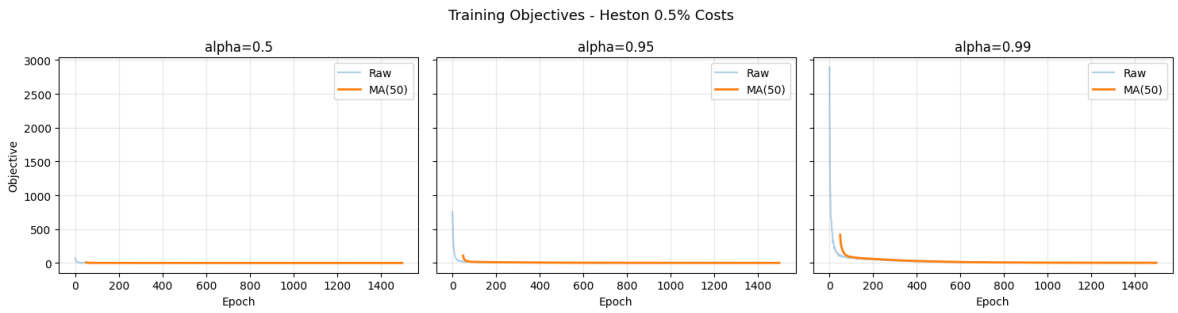


(a) Left-tail zoom of empirical CDF (bottom 20%).

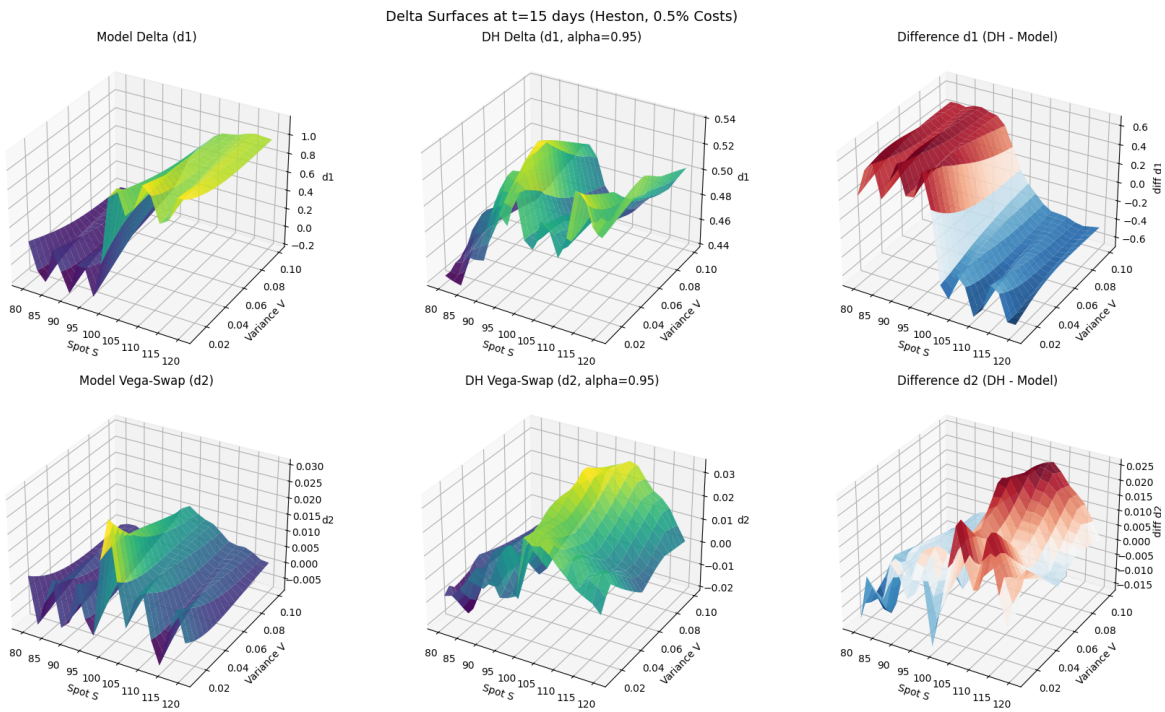


(b) P&L distributions across all risk levels.

Figure 15: Tail analysis and multi-level comparison, Heston with transaction costs ($\epsilon = 0.005$).



(a) Training objectives (raw and MA(50)) for all α levels.



(b) Delta surfaces at $t = 15$ days: model vs. DH ($\alpha = 0.95$) with difference.

Figure 16: Training diagnostics and delta surfaces, Heston with transaction costs ($\epsilon = 0.005$).

The P&L histogram (Figure 14a) confirms the central result: the deep hedge distribution is shifted substantially to the right of the model delta baseline (-0.46 vs. -1.17 mean), with the left tail truncated significantly. The full empirical CDF (Figure 14b) reveals a crossover around -1.5: to the left of this point the deep hedge has strictly less probability mass than the baseline, reflecting fewer extreme losses, while to the right it has a fatter tail from paths where the network under-hedges and the option expires at low cost. This is consistent with the CVaR improvements reported in Table 9 alongside the higher standard deviation.

The left-tail zoom (Figure 15a) makes the tail improvement particularly clear: at a loss level of -3, the model delta baseline accumulates roughly three times more probability mass than the deep hedge. The multi-level histogram (Figure 15b) visually confirms the degenerate behaviour of $\alpha = 0.5$: its distribution is wide and flat, extending beyond -20, in sharp contrast to the $\alpha = 0.95$ and $\alpha = 0.99$ strategies which are nearly identical and consistently shifted right of the baseline.

Training curves (Figure 16a) show clean convergence for all three risk levels, with the initial objective scaling from approximately 300 at $\alpha = 0.5$ to approximately 2700 at $\alpha = 0.99$, reflecting the increasing weight assigned to extreme tail losses. Convergence is achieved within the first 200 epochs; the 1500 epoch budget is comfortably sufficient.

The delta surface comparison (Figure 16b) reveals the learned hedging behaviour at $t = 15$ days. The model d1 surface exhibits the expected S-curve shape across spot prices, ranging from near zero for deep OTM to near one for deep ITM. The DH d1 surface is markedly flatter (0.44 to 0.54), indicating that the network systematically reduces its stock position relative to the model hedge to save transaction costs. The difference surface confirms that the DH strategy hedges less aggressively for ITM positions and more aggressively for OTM positions, effectively flattening the delta profile. For d2, both model and network operate on a similar scale (0 to 0.03), with noisy differences reflecting the network's greater uncertainty in hedging the variance dimension.

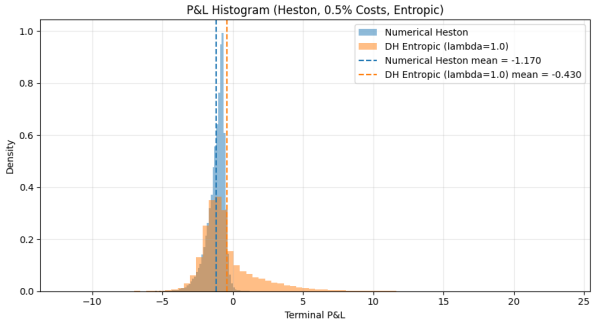
4.2.3 Heston: Entropic Risk Measure

Table 10: Heston, $\epsilon = 0.005$: three-way comparison between the numerical baseline, Deep Hedge CVaR ($\alpha = 0.95$) and Deep Hedge Entropic ($\lambda = 1$). Entropic model trained in 69.2s on NVIDIA RTX 5070 Ti (laptop), 1500 epochs, batch size 4096.

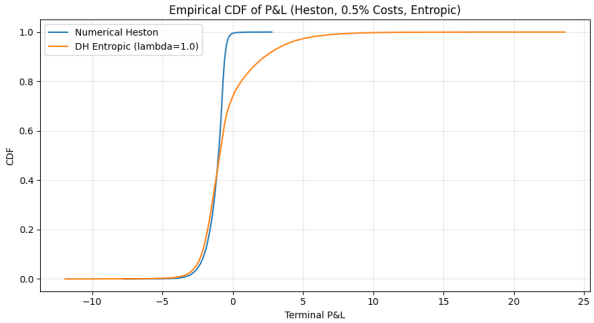
Metric	**Heston Baseline**	**DH CVaR** $\alpha = 0.95$	**DH Entropic** $\lambda = 1$
Mean P&L	-1.1703	-0.4556	-0.4299
Std P&L	0.6463	1.4180	2.0711
VaR (95%)	2.4216	1.9153	2.6386
CVaR (95%)	2.9688	2.3763	3.3345
VaR (99%)	3.2842	2.6302	3.6903
CVaR (99%)	3.8624	3.3057	4.6023
Turnover	2.2579	0.8537	0.8148
Avg Transaction Cost	1.1243	0.4231	0.4044

Table 10 reveals an important divergence from the Black & Scholes results. In the Heston setting, the entropic objective ($\lambda = 1$) achieves the best mean P&L (-0.43) and lowest average transaction costs (0.40), consistent with its cost-minimising behaviour observed in Black & Scholes. However, unlike in the simpler setting, its tail metrics are now worse than the numerical baseline: CVaR(95%) rises to 3.33 compared to the baseline's 2.97, and CVaR(99%) to 4.60 versus 3.86. This is a consequence of the extremely high variance (2.07) induced by the fat right tail visible in Figure 17a: the network learns to under-hedge systematically, retaining the premium on paths where the option expires worthless but suffering catastrophic losses on adverse paths. The exponential utility with $\lambda = 1$ is insufficiently risk-averse to prevent this behaviour under the additional uncertainty introduced by stochastic volatility. The CVaR objective at $\alpha = 0.95$ remains the clear choice for tail risk management in the Heston setting, achieving a 20% reduction in CVaR(95%) while the entropic objective provides no tail improvement whatsoever and in fact degrades tail performance relative to the classical baseline.

Figures

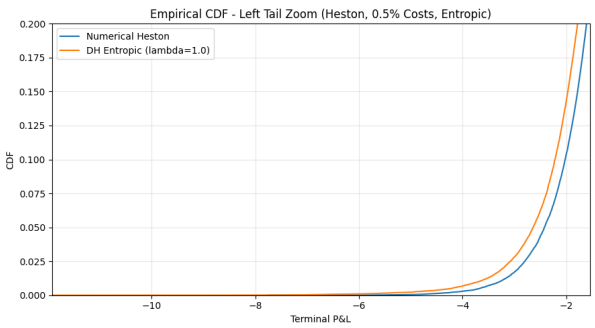


(a) P&L histogram: Heston baseline vs. Deep Hedge Entropic ($\lambda = 1$).

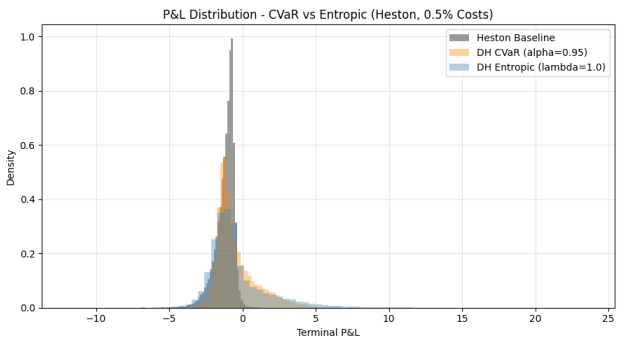


(b) Empirical CDF of terminal P&L.

Figure 17: P&L distribution comparison, Heston with transaction costs ($\epsilon = 0.005$), Entropic objective ($\lambda = 1$).



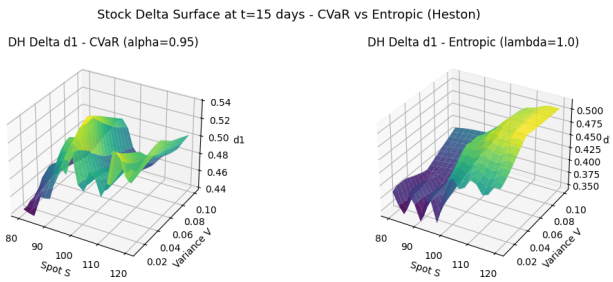
(a) Left-tail zoom of empirical CDF (bottom 20%).



(b) P&L distributions: baseline vs. CVaR vs. Entropic.

Figure 18: Tail analysis and three-way comparison, Heston with transaction costs ($\epsilon = 0.005$).

(a) Training objective (Entropic risk, raw and MA(50)).



(b) Stock delta surface: CVaR vs. Entropic at $t = 15$ days.

Figure 19: Training diagnostics and delta comparison, Heston Entropic objective ($\epsilon = 0.005$).

The histogram (Figure 17a) makes the entropic failure mode immediately visible: the distribution has an extreme right tail extending beyond +20, reflecting paths where the network

barely hedges and profits from option expiry. The full CDF (Figure 17b) shows that in the moderate loss region both strategies track closely, but the left-tail zoom (Figure 18a) reveals that at the extreme left the entropic curve lies above the baseline, confirming that it accumulates more extreme losses. The three-way overlay (Figure 18b) makes the advantage of CVaR clear: its distribution is tighter and shifted right while the entropic distribution straddles the baseline on both sides.

The training curve (Figure 19a) shows a turbulent convergence with an initial spike near 4000 and several secondary spikes through epoch 600, reflecting the difficulty of optimising the exponential objective under the high-variance Heston dynamics. Convergence is eventually achieved, but less smoothly than in the CVaR case. The delta surface comparison (Figure 19b) shows that both learned strategies are much flatter than the analytical model delta. The entropic surface (0.35 to 0.50) hedges slightly less aggressively than the CVaR surface (0.44 to 0.54), consistent with its lower turnover, and shows a stronger dependence on variance V at low spot values, reflecting the network's attempt to adapt to volatility regime changes.

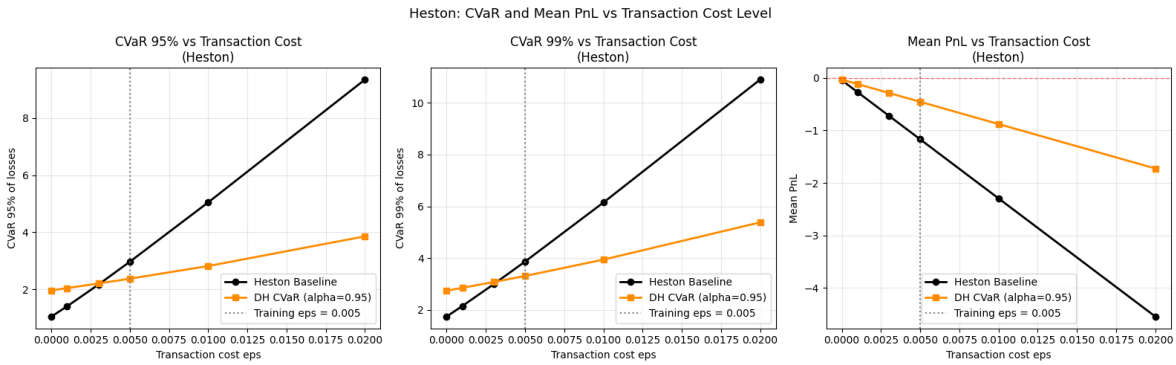
4.2.4 Heston CVaR vs Transaction Cost Level

Table 11: Heston CVaR(95%), CVaR(99%) and Mean P&L as a function of transaction cost level ϵ. The DH model is fixed at the policy trained at $\epsilon = 0.005$, $\alpha = 0.95$, and is not retrained at each cost level.

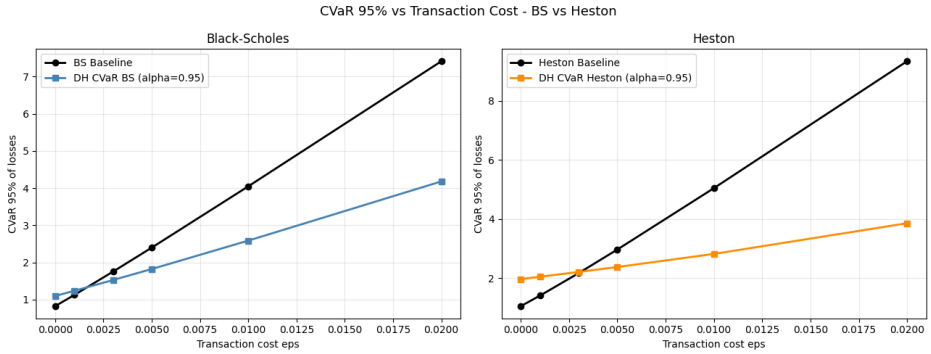
ϵ	Base CVaR(95%)	DH CVaR(95%)	Base CVaR(99%)	DH CVaR(99%)	Base Mean P&L	DH Mean P&L
0.0000	1.0518	1.9673	1.7371	2.7379	−0.0460	−0.0325
0.0010	1.4099	2.0471	2.1444	2.8470	−0.2708	−0.1171
0.0030	2.1705	2.2093	2.9874	3.0724	−0.7206	−0.2863
0.0050	2.9688	2.3763	3.8624	3.3057	−1.1703	−0.4556
0.0100	5.0468	2.8193	6.1465	3.9384	−2.2945	−0.8787
0.0200	9.3436	3.8579	10.9022	5.3784	−4.5431	−1.7248

Table 11 and Figure 20a confirm the robustness of the deep hedge advantage across a wide range of transaction cost levels. At $\epsilon = 0$ the baseline achieves lower CVaR, as the numerical Heston delta is near-optimal in the frictionless setting. The crossover occurs between $\epsilon = 0.003$ and $\epsilon = 0.005$: at $\epsilon = 0.003$ the two strategies are nearly indistinguishable (CVaR(95%) 2.17 vs. 2.21), while at $\epsilon = 0.005$ the deep hedge clearly dominates (2.97 vs. 2.38). Beyond the training cost level the advantage grows dramatically: at $\epsilon = 0.02$ the baseline CVaR(95%) reaches 9.34 while the deep hedge remains at 3.86, a reduction of 59%. Mean P&L shows the same pattern: the baseline degrades almost linearly to -4.54 while the deep hedge retains -1.72 at $\epsilon = 0.02$. Since the DH model is evaluated at its fixed trained policy rather than retrained at each cost level, a model specifically retrained at each ϵ would show an even stronger advantage.

The comparison plot (Figure 20b) shows that the qualitative behaviour is identical across Black & Scholes and Heston: baselines degrade linearly with ϵ while deep hedges flatten. The Heston deep hedge starts at a higher absolute CVaR level (approximately 2.0 at $\epsilon = 0$) compared to the Black & Scholes case (approximately 1.0), reflecting the additional unhedgeable risk from stochastic volatility, but the relative advantage over the classical baseline is equally pronounced in both settings.



(a) Heston: CVaR(95%), CVaR(99%) and Mean P&L vs. ϵ. Dotted line marks the training cost level $\epsilon = 0.005$.



(b) CVaR(95%) vs. ϵ: Black & Scholes and Heston side by side.

Figure 20: CVaR as a function of transaction cost level, Heston ($\alpha = 0.95$).

4.2.5 Hyperparameter Sensitivity: Heston

Table 12: Heston network width sensitivity: CVaR metrics, turnover and training time at $\alpha = 0.95$, $\epsilon = 0.005$. All runs on NVIDIA RTX 5070 Ti (laptop), 1500 epochs, batch size 4096.

Config	Mean P&L	Std P&L	CVaR(95%)	CVaR(99%)	Turnover	Time (s)
Small $[8, 8]$	-0.4356	1.6720	2.8448	4.2331	0.8100	72.1
Paper $[17, 17]$	-0.4547	1.5128	2.4237	3.3395	0.8556	72.0
Large $[128, 128]$	-0.6120	0.8277	1.6178	1.8891	1.1551	71.2

Table 13: Heston batch size sensitivity: CVaR metrics, turnover and training time at $\alpha = 0.95$, $\epsilon = 0.005$, hidden dims $[17, 17]$. All runs on NVIDIA RTX 5070 Ti (laptop), 1500 epochs.

Config	**Mean P&L**	**Std P&L**	**CVaR(95%)**	**CVaR(99%)**	**Turnover**	**Time (s)**
Batch 64	-0.3031	1.7977	4.2766	7.1320	0.5349	42.1
Batch 256	-0.4298	1.7845	3.3227	5.1861	0.7911	42.1
Batch 512	-0.4184	1.8003	3.0449	4.5785	0.7706	42.5
Batch 4096	-0.4571	1.4199	2.3909	3.2788	0.8574	72.1

Table 12 reveals a more pronounced width effect than in the Black & Scholes setting. While the paper's $[17, 17]$ architecture meaningfully outperforms $[8, 8]$ (CVaR(95%) 2.42 vs. 2.84), the large $[128, 128]$ network provides a substantial further improvement (1.62), unlike in Black & Scholes where diminishing returns set in at $[32, 32]$. This suggests that the two-asset stochastic volatility problem genuinely benefits from additional network capacity, consistent with the higher complexity of jointly learning stock and variance-swap hedging strategies. Training time remains stable at approximately 71 to 72 seconds across all widths, confirming that width has negligible impact on computational cost in this setting.

Table 13 shows a strong monotone improvement across all batch sizes, more pronounced than in Black & Scholes. The improvement from batch 512 to 4096 is substantial: CVaR(95%) falls from 3.04 to 2.39 and CVaR(99%) from 4.58 to 3.28, unlike in Black & Scholes where this step was marginal. This strongly justifies the choice of batch size 4096 in the Heston experiments: larger batches provide more stable gradient estimates for the two-asset on-policy training loop, where path-to-path variance is significantly higher than under constant volatility. The trade-off is a modest increase in training time from 42 to 43 seconds at small batches to 72.1 seconds at batch 4096, which remains negligible in absolute terms.

Figures

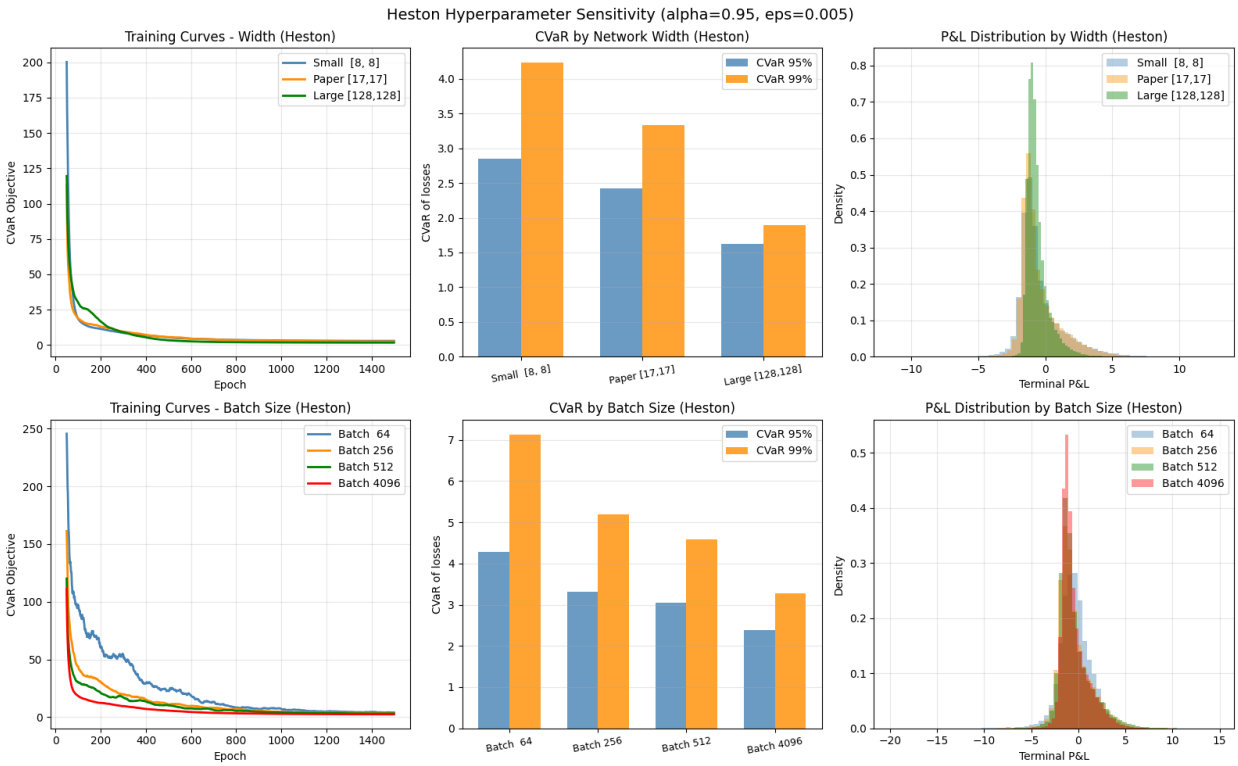


Figure 21: Heston hyperparameter sensitivity ($\alpha = 0.95$, $\epsilon = 0.005$). Top row: network width, showing training curves (left), CVaR bar chart (centre), and P&L distributions (right). Bottom row: batch size, same layout

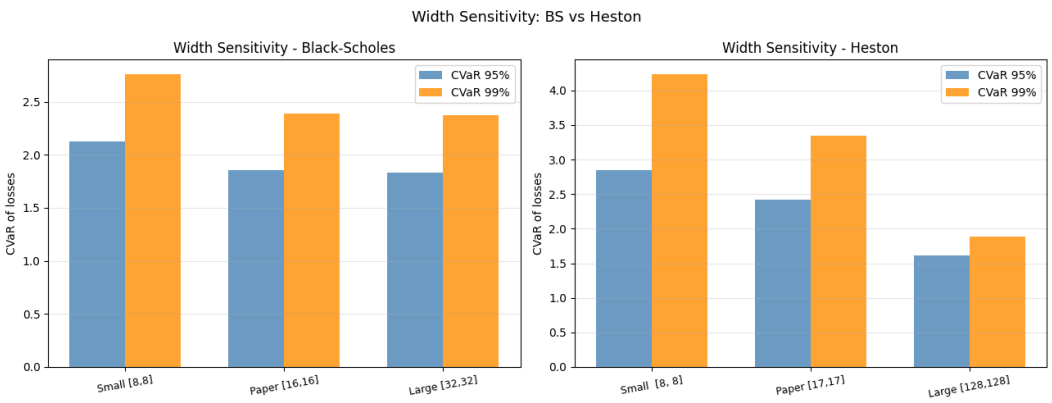


Figure 22: Width sensitivity comparison: Black & Scholes (left) vs. Heston (right). CVaR(95%) and CVaR(99%) by network width. In Black & Scholes, diminishing returns set in after $[16, 16]$; in Heston, performance continues to improve through $[128, 128]$.

The six-panel figure (Figure 21) confirms the findings from Tables 12 and 13. For network width, the P&L distribution panel clearly shows the Large $[128, 128]$ network producing a substantially tighter distribution with fewer extreme losses, while Small $[8, 8]$ exhibits the heaviest tails. Batch size 64 produces the most dispersed distribution, with tails extending

beyond 15 in absolute value, visually confirming the gradient noise problem at small batch sizes. The training curve for Batch 64 also fails to fully stabilise by epoch 1500, further justifying the use of larger batches.

The comparison plot (Figure 22) highlights the key structural difference between the two environments: in Black & Scholes the CVaR improvement saturates after the paper's [16, 16] architecture, while in Heston it continues to decrease through [128, 128]. This confirms that the stochastic volatility setting presents a strictly harder approximation problem, requiring greater network capacity to represent the joint stock and variance-swap hedging policy.

4.3 Price Asymptotics Under Proportional Transaction Costs

4.3.1 Black & Scholes

A key theoretical result of Buehler et al. (2019), originally established by Whalley and Wilmott (1997) and Kallsen and Muhle-Karbe (2015), is that the exponential utility indifference price p_ϵ satisfies the asymptotic relation

$$p_\epsilon - p_0 = O(\epsilon^{2/3}), \quad \epsilon \downarrow 0, \tag{30}$$

where p_0 is the risk-neutral price. We verify this numerically using the deep hedge entropic objective with $\lambda = 1$. For each ϵ in the sequence $\varepsilon_i = 2^{-i+5}$, $i = 1, \ldots, 5$, we train two models with fresh random seeds at each epsilon level: one with the option liability included to approximate $\pi(-Z; \epsilon)$ and one without to approximate $\pi(0; \epsilon)$. The indifference price is then recovered as $p_\epsilon = \pi(-Z; \epsilon) - \pi(0; \epsilon)$ via equation (3.2) of Buehler et al. (2019).

Table 14: Black & Scholes price asymptotics: indifference price p_ϵ and excess $p_\epsilon - p_0$ for the epsilon sequence $\varepsilon_i = 2^{-i+5}$. Risk-neutral price $p_0 = 2.2871$, entropic objective $\lambda = 1$, 500 epochs, batch size 4096, NVIDIA RTX 5070 Ti (laptop).

ϵ	p_ϵ	$p_\epsilon - p_0$	$\log(\epsilon)$
0.01562	4.1425	1.8554	−4.158
0.00781	3.3766	1.0894	−4.852
0.00391	2.9532	0.6661	−5.543
0.00195	2.7067	0.4196	−6.236
0.00098	2.5584	0.2712	−6.929

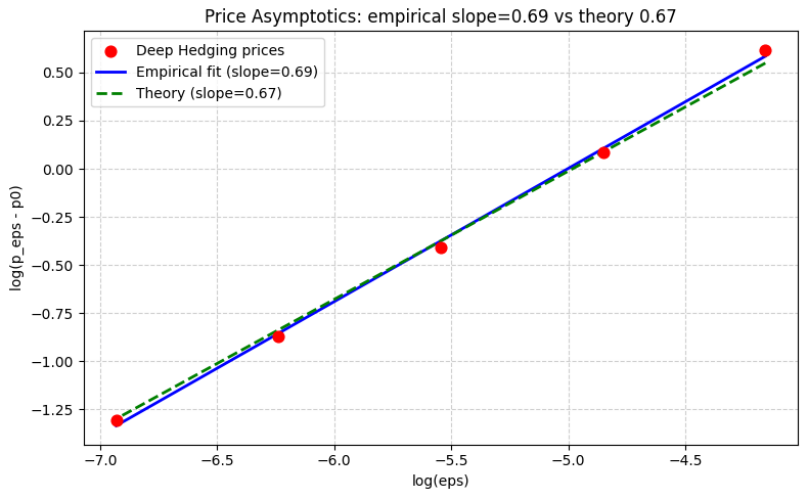


Figure 23: Black & Scholes price asymptotics: $\log(p_\epsilon - p_0)$ vs. $\log(\epsilon)$. Empirical slope 0.69 (blue) vs. theoretical slope $2/3 \approx 0.67$ (green dashed).

Table 14 and Figure 23 confirm the $O(\epsilon^{2/3})$ relation. The log-log plot shows all five price differences lying almost perfectly on a straight line, with an empirical OLS slope of 0.692, extremely close to the theoretical value of $2/3 \approx 0.667$. The $p_\epsilon - p_0$ values decrease monotonically from 1.86 at $\epsilon = 0.016$ to 0.27 at $\epsilon = 0.001$, confirming that the network correctly estimates the indifference price at each cost level. The near-perfect alignment of the empirical fit with the theoretical reference line replicates Figure 10 of Buehler et al. (2019) and confirms that the deep hedging algorithm correctly recovers the known asymptotic behaviour of the exponential utility indifference price under proportional transaction costs in the Black & Scholes model.

4.3.2 Heston

We follow the same procedure as for Black & Scholes: for each ϵ in the sequence $\varepsilon_i = 2^{-i-3}$, $i = 1, \ldots, 5$, we train two entropic models with fresh random seeds at each epsilon level, one with the option liability included to approximate $\pi(-Z; \epsilon)$ and one without to approximate $\pi(0; \epsilon)$, and compute $p_\epsilon = \pi(-Z; \epsilon) - \pi(0; \epsilon)$ via equation (3.2) of Buehler et al. (2019). Training uses 1000 epochs and batch size 1024, with hidden dimensions $[64, 64]$ to provide sufficient capacity for accurate indifference price estimation at each epsilon level.

Table 15: Heston price asymptotics: indifference price p_ϵ and excess $p_\epsilon - p_0$ for five epsilon values. Risk-neutral price $p_0 = 1.6444$, entropic objective $\lambda = 1$, 1000 epochs, batch size 1024, hidden dims $[64, 64]$, NVIDIA RTX 5070 Ti (laptop).

ϵ	p_ϵ	$p_\epsilon - p_0$	$\log(\epsilon)$	$\log(p_\epsilon - p_0)$
0.01562	8.8138	7.1695	−4.159	1.970
0.00781	6.7970	5.1526	−4.852	1.639
0.00391	3.6900	2.0456	−5.545	0.716
0.00195	3.2376	1.5932	−6.238	0.466
0.00098	3.0990	1.4546	−6.931	0.375

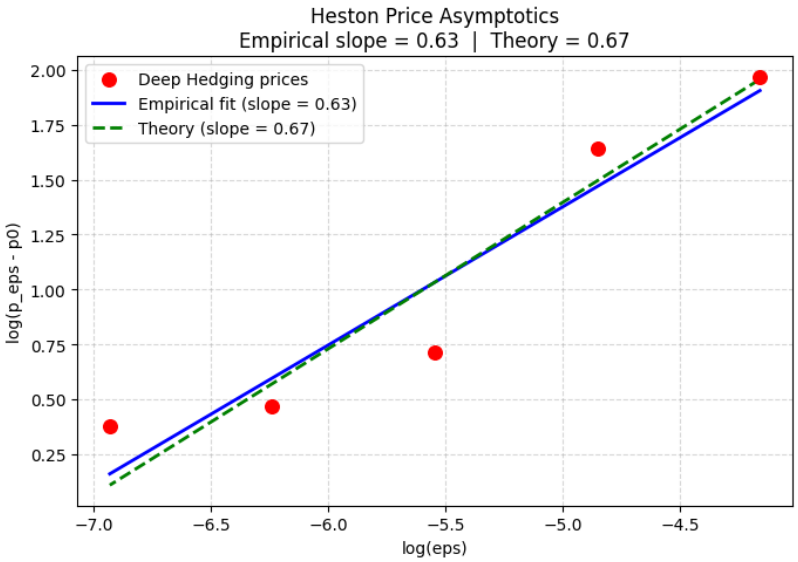


Figure 24: Heston price asymptotics: $\log(p_\epsilon - p_0)$ vs. $\log(\epsilon)$. Empirical slope 0.63 (blue) vs. theoretical slope $2/3 \approx 0.67$ (green dashed).

Table 15 and Figure 24 show that the $O(\epsilon^{2/3})$ qualitative behaviour is confirmed in the Heston two-asset setting, with an empirical OLS slope of 0.630. The $p_\epsilon - p_0$ values decrease broadly from 7.17 at $\epsilon = 0.016$ to 1.45 at $\epsilon = 0.001$, consistent with the predicted scaling. The deviation of the empirical slope from the theoretical value of $2/3 \approx 0.667$ is attributable to two factors. First, the two smallest epsilon points ($\epsilon = 0.00195$ and $\epsilon = 0.00098$) exhibit similar $p_\epsilon - p_0$ values (1.59 and 1.45) despite a factor of two difference in ϵ, suggesting that at these cost levels the indifference price difference approaches the noise floor of the training procedure and cannot be estimated reliably with 1000 epochs and batch size 1024. Second, the on-policy training under the Heston model introduces higher path-to-path variance than in the Black & Scholes setting, making the entropic objective harder to optimise at small epsilon levels where the signal in $p_\epsilon - p_0$ is small. These limitations are noted as directions for future work in Section 5. Nevertheless, the positive slope and the broadly decreasing trend in the log-log plot confirm the qualitative $O(\epsilon^{2/3})$ behaviour, extending the numerical

evidence of Buehler et al. (2019) to the two-asset stochastic volatility setting.

5 Conclusion

This report presented a full replication and extended analysis of the Deep Hedging framework of Buehler et al. (2019), implemented from scratch in PyTorch across two market environments, Black & Scholes and Heston stochastic volatility, and two risk objectives, Conditional Value-at-Risk and the entropic risk measure.

When Deep Hedging Succeeds

The central finding of Buehler et al. (2019) is confirmed: deep hedging provides substantial and consistent improvements over classical model-based hedging once proportional transaction costs are introduced. In the Black & Scholes setting, the tail-focused deep hedge at $\alpha = 0.95$ reduces CVaR(95%) by 24% (from 2.3970 to 1.8206) and average transaction costs by 42% (from 1.1116 to 0.6459) relative to the cost-oblivious analytical delta. In the Heston setting, the corresponding reductions are 20% in CVaR(95%) (from 2.9688 to 2.3763) and 62% in average transaction costs (from 1.1243 to 0.4231). These improvements are robust across a wide range of transaction cost levels beyond the training value of $\epsilon = 0.005$: at $\epsilon = 0.02$ the Heston baseline CVaR(95%) reaches 9.34 while the deep hedge remains at 3.86, a reduction of 59%. The learned policy internalises the cost of rebalancing and adapts by trading less aggressively, with turnover falling from 2.22 to 1.29 in Black & Scholes and from 2.26 to 0.85 in Heston.

The improvements are achieved with training times that are competitive with or faster than computing the classical numerical baseline. In the Heston setting, evaluating the Fourier-based greeks across the full 50,000-path test set requires 114.5 seconds, while training the deep hedge requires approximately 67 to 70 seconds on an NVIDIA RTX 5070 Ti laptop GPU. This confirms the scalability argument of Buehler et al. (2019): the computational cost of the deep hedge depends primarily on the number of hedging instruments rather than on the complexity of the derivatives portfolio.

When Deep Hedging Fails or Requires Care

Two failure modes were identified. First, mild risk objectives such as CVaR at $\alpha = 0.5$ are insufficient to prevent the network from discovering a degenerate solution in the Heston setting: with only 50% of paths contributing to the CVaR gradient, the penalty on tail losses is too weak to overcome the incentive to reduce transaction costs by simply not hedging. The resulting strategy collapses to a near-zero turnover of 0.35, achieving a better mean

P&L of -0.22 but generating catastrophic tail losses with CVaR(95%) of 4.71, far worse than the model delta baseline of 2.97. This failure mode does not arise in Black & Scholes, where the simpler one-dimensional problem provides a stronger gradient signal. Tail-focused objectives at $\alpha = 0.95$ or $\alpha = 0.99$ consistently avoid this issue.

Second, the entropic risk measure with $\lambda = 1$ proves insufficiently risk-averse in the Heston setting. While it achieves the best mean P&L (-0.43) and lowest transaction costs (0.40) in the Heston environment by learning to trade conservatively, its CVaR(95%) of 3.33 is worse than the classical baseline of 2.97, due to the high variance of 2.07 induced by systematic under-hedging under stochastic volatility. In Black & Scholes the entropic model remains competitive, achieving CVaR(95%) of 1.99 versus the CVaR model's 1.82. The CVaR objective is therefore the clear recommendation for practitioners who prioritise tail risk management in stochastic volatility environments, while the entropic objective may be preferred when cost minimisation is the primary concern in simpler settings.

Capacity Requirements and Environment Complexity

The hyperparameter sensitivity analysis reveals a fundamental difference between the two environments. In Black & Scholes, the paper's default architecture of two hidden layers with $d + 15$ nodes reaches near-optimal performance: increasing from $[16, 16]$ to $[32, 32]$ reduces CVaR(95%) only marginally from 1.85 to 1.83, and larger batch sizes beyond 512 provide negligible further improvement. In Heston, performance continues to improve substantially with larger networks: the $[128, 128]$ architecture achieves CVaR(95%) of 1.62 compared to 2.42 for the paper's $[17, 17]$ design, a reduction of 33%. Batch size similarly has a more pronounced effect in Heston: CVaR(95%) falls from 4.28 at batch 64 to 2.39 at batch 4096, while the corresponding reduction in Black & Scholes is from 2.42 to 1.83. This reflects the genuinely higher complexity of jointly learning stock and variance-swap hedging strategies under stochastic volatility, and suggests that production deployments in realistic multi-asset settings should allocate more network capacity and larger batch sizes than the paper's original prescription.

Price Asymptotics

The theoretical $O(\epsilon^{2/3})$ scaling of the exponential utility indifference price under proportional transaction costs is confirmed numerically in both environments. In Black & Scholes, the empirical OLS slope of 0.692 is in close agreement with the theoretical value of $2/3 \approx 0.667$, with all five log-log points lying nearly perfectly on a straight line. In the Heston two-asset setting, the empirical slope of 0.630 confirms the qualitative $O(\epsilon^{2/3})$ behaviour, though with greater deviation from the theoretical value due to training noise at small epsilon levels where the indifference price difference is difficult to estimate reliably with the available

computational budget.

Limitations and Future Work

Several limitations of the present study should be noted. The Heston path simulator uses the trapezoidal approximation for integrated variance rather than the exact Broadie-Kaya scheme, introducing a small bias in the simulated price distribution and a slight discrepancy in the option premium (1.6444 vs. the paper's 1.69). Batch normalisation, applied in the original paper, was disabled throughout due to training instability under on-policy simulation, and a more systematic investigation of this choice would be valuable. The asymptotic experiment in the Heston setting is sensitive to the training budget and network capacity: the two smallest epsilon points exhibit similar $p_\epsilon - p_0$ values suggesting the signal approaches the noise floor of the estimation procedure, and a larger epoch count or more runs with different seeds could improve the reliability of the empirical slope estimate.

Future work could extend the analysis to more realistic market environments, including multiple underlyings, path-dependent payoffs, or empirically calibrated market dynamics, where the model-free nature of deep hedging provides its greatest advantage over classical approaches. The comparison between per-step and shared policy architectures across different payoff complexities would also be a natural extension of the sensitivity analysis presented here.

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